



An ISO 9001:2015 Company

Technical Handbook



Uncompromised Engineering

About Us

Born of entrepreneurial spirit in 2014. Q-Tork carved an ornament for itself as a pioneer in the design and manufacturing of gear operators in India.

Head office based in Pune has started its journey from a single product, single product series and with a single customer in a rented premises at Kuruli industrial area near Chakan with a very limited staff and machineries.

Q-Tork presently having multi products, multi product series with very good range of products in all series and multiple customers all over India and overseas too.

Company has 1800 Sq.Mtr. area for the manufacturing Products as per the Customer requirements. Company has trained service persons to carry out different operations. Q-Tork with a compound annual growth of 45%+ has one of the largest manufacturing capacity of 500+ gear operators in a day with best manufacturing facility along with very good control on all processes of gear operator manufacturing.

The gear operators can also be installed on the valve depend upon the size of the valve it will be carried out on request.

Engineering Excellence

Q-Tork has a dedicated Engineering Development Centre with pool of experienced and skilled engineers. This team constantly strives to update their knowledge on gear engineering and valves, using latest software - 2D, 3D and Finite Element Analysis (FEA).

New product development or upgrade in product happens with utmost Uncompromised Engineering that passes through phases like – Dynamic test, Static test, Ingress Protection (IP) test and so on to ensure customer satisfaction of highest level.

Manufacturing

Q-Tork is committed to being a world class organization by manufacturing gear operators to domestic and overseas. It has the state of art manufacturing facility located in Markal, Pune, Maharashtra, India.

Q-Tork also adopts stringent quality system to develop and manufacturing of world class products in its plant. Critical processes like whirling, broaching, assembly and Product testings are taken care in-house

Q-Tork also excels in producing of specialized or all types of optional material for a small and large quantity.

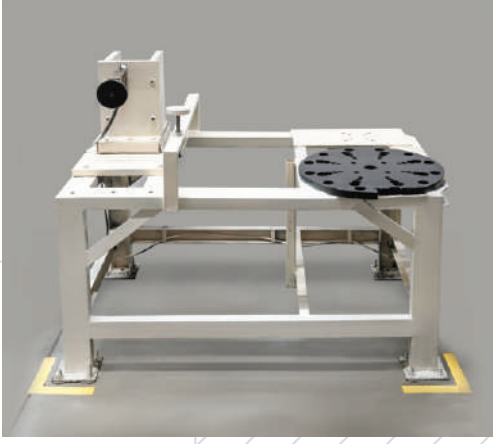
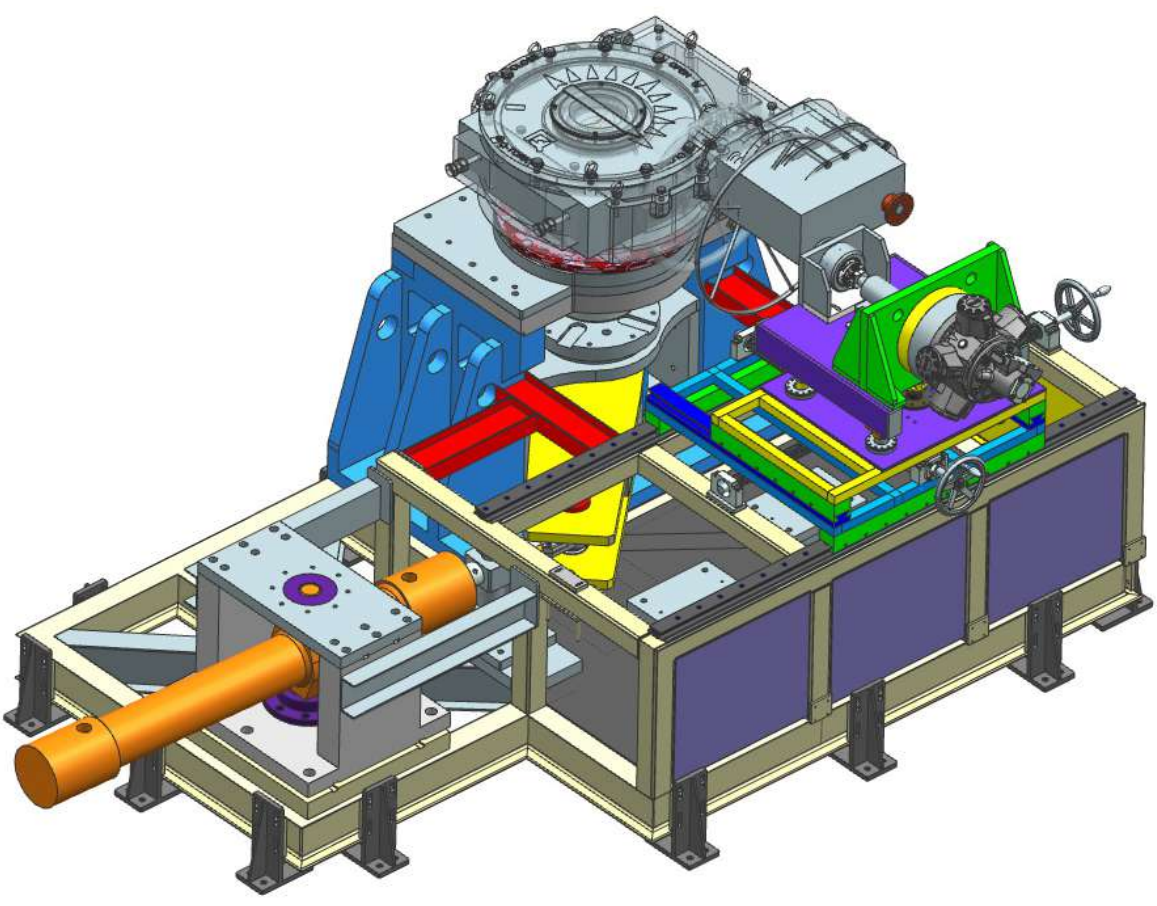
Quality Assurance

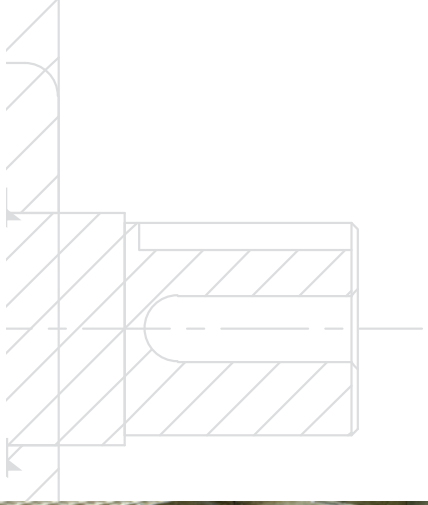
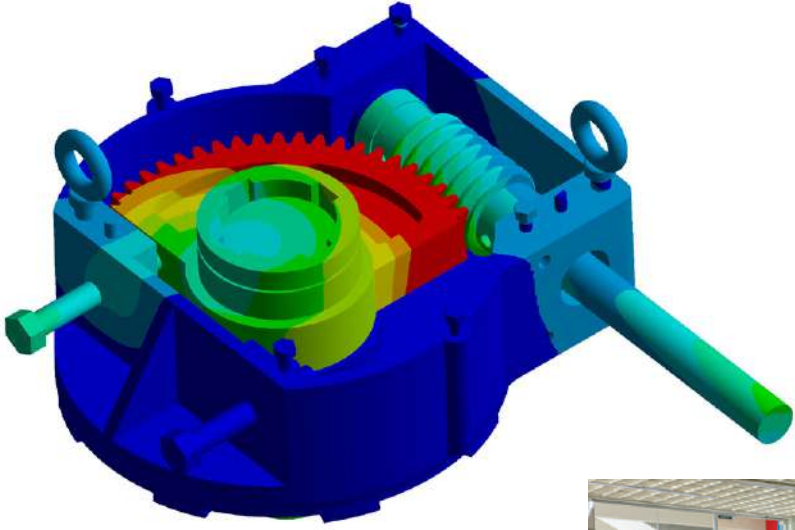
Q-Tork is committed to being a World Class Organization supplies gear operators and its components to domestic and overseas both sectors. It strives to comply with the highest possible quality standards and practices a philosophy of continuous improvement. It adopts processes that ensure full proof and defect free manufacturing. Q-Tork also seamlessly and efficiently satisfies its customer needs by optimizing production cycles. Continuous improvement through Kaizen, lean manufacturing practices, employee training, suggestion schemes and environment management systems are the hallmarks of Q-Tork's success in customer retention and growth. Its focused involvement in all the processes further ensure customer satisfaction and superior product quality. we have developed a reputation for high quality with dependable gearbox operators.













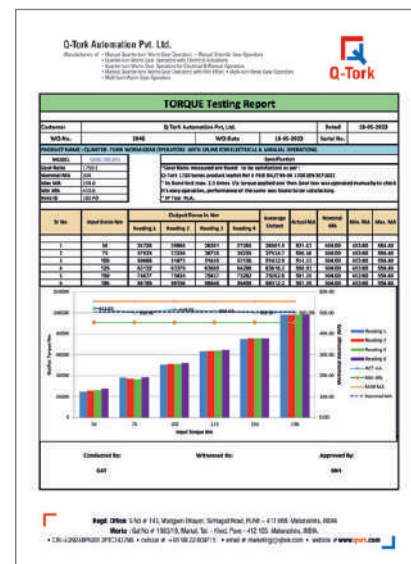
■ ISO Certificate



■ CE Certificate



■ MSME ZED Bronze Certificate



■ Torque Testing Report



■ IP68 Certificate



■ PoD Certificate

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311 Series | Manual Quarter Turn Worm Gear Operators

Q-Tork 311 series manual operated gear boxes has been designed keeping in mind the needs of durability and performance. To ensure stringent quality standards and testing norms.

Application

311 series quarter turn worm gearboxes are used to operate Ball valve, Butterfly valve, Plug valve and Dampers.

Features & Benefits

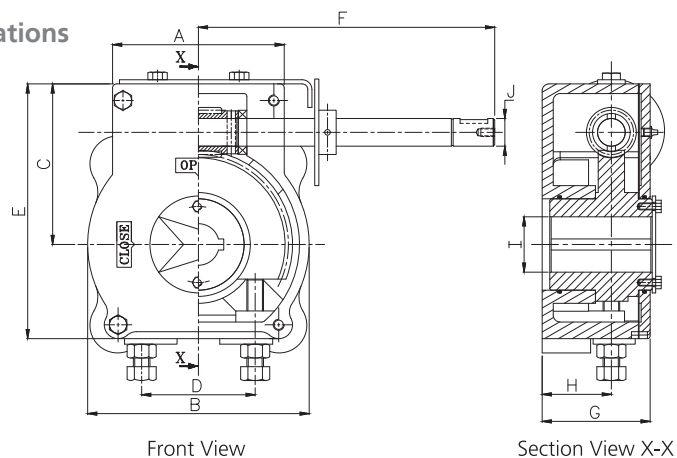
- Graded CI Housing & Cover
- Comprehensive gear ratios with spur input reduction section
- High performance axial bearing
- Powder coated hand wheel for easy and smooth operation
- Temperature range -30°C to $+120^{\circ}\text{C}$
- Light weight with rugged design
- Maintenance Free
- Environmental friendly high performance grease
- Alloy steel-worm & input shaft

Optional

- Offering MOC:
 - ▶ Housing/ Cover- WCB, LCB, LCC, Ductile Iron, Stainless Steel
 - ▶ Quadrant - Al. Bronze
- Stainless steel- Input shaft, handwheel and hardware
- Lubricants in temperature variants can be provided.
- Submersibility to IP67
- Vertical operation with bevel gear attachment
- Locking arrangement
- Chain wheel operation
- Limit switch arrangement

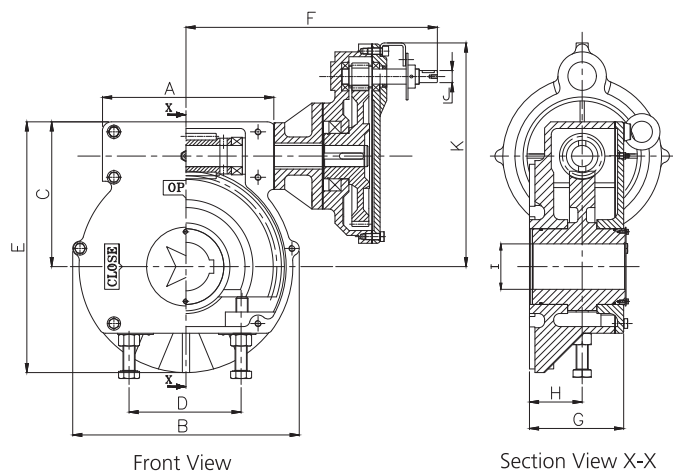
■ Dimensional Specifications

■ FIG (1)



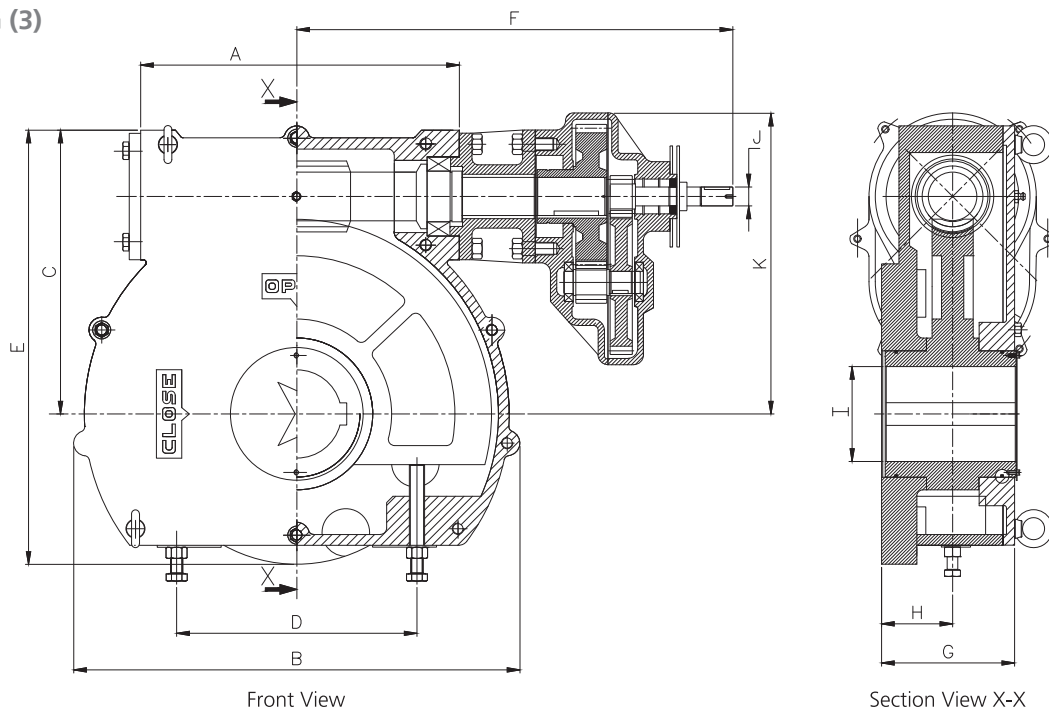
Sr. No.	Models	A	B	C	D	E	F	G	H	I	J
1	QTA-150	72	72	54	37	81	111	55	34	20	12
2	QTA-250	90	90	66	58	111	161.5	53.5	27.5	22	12
3	QTA-500	110	110	81	64	121	153	60	33	35	15
4	QTA-750	104	116	91	68	135	161.5	64	36	35	20
5	QTA-1000	116	160	106	84	166	218.5	81	48	45	20
6	QTA-1250	124	160	116	86	184	214	78	50	45	20
7	QTA-1500	139	191	130	91	203	214	88.5	55	50	20
8	QTA-2000	172	197	135	108	213	251	90.5	54	70	20
9	QTA-3000	164	218	152	110	232	255	90	54.5	70	20
10	QTA-3750	185	300	171	146	261	255.5	108	60	70	20
11	QTA-4000	213	300	195	150	344	250	119	71	85	20
12	QTA-4500	213	300	195	150	344	250	119	71	85	20

■ FIG (2)



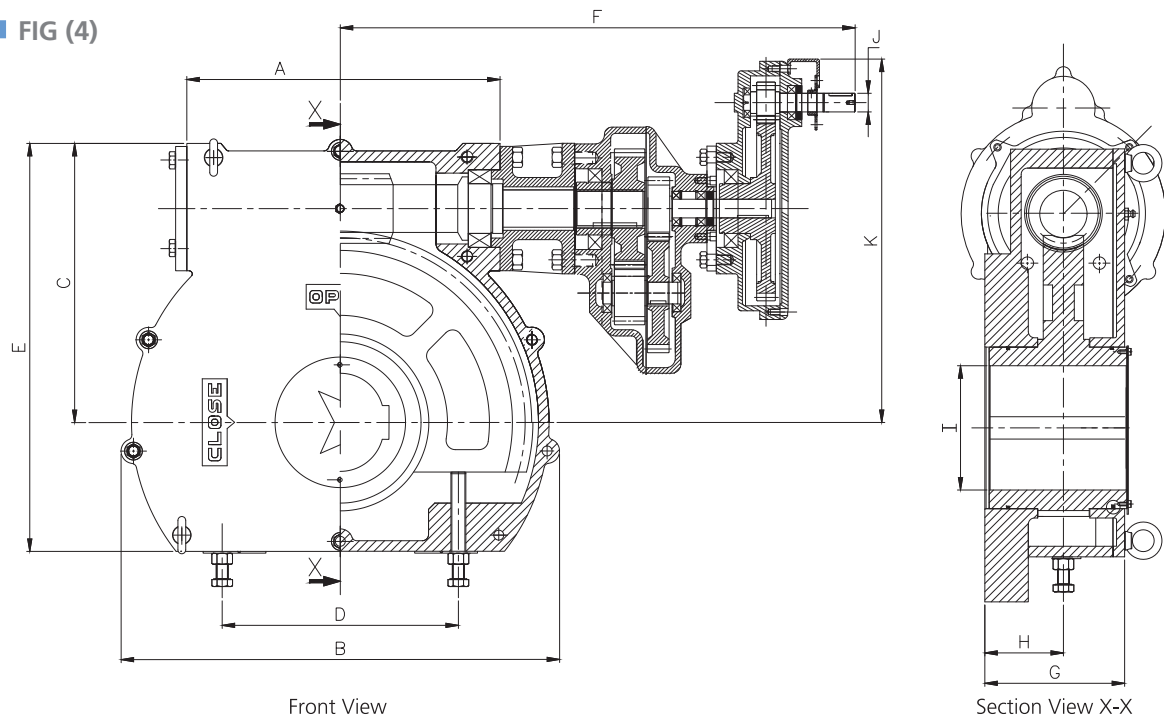
Sr.No.	MODELS	A	B	C	D	E	F	G	H	I	J	K
13	QTA-5500 / SR1	250	300	195	150	345	300	119	70.5	85	20	247
14	QTA-6000 / SR1	250	300	195	150	345	300	119	70.5	85	20	247
15	QTA-7000 / SR1	254	350	202	146	377	285	147	91.5	95	20	251.5
16	QTA-8500 / SR2	257	350	220	170	412	285.5	147	89.5	95	20	281
17	QTA-10000 / SR2	279	334	237	170	346	307	154	96.5	95	20	286
18	QTA-13000 / SR3	280	353	239	185	415	415	156	87	95	20	308
19	QTA-16000 / SR3	305	410	268	234	443	428	162	96	110	20	322
20	QTA-18500 / SR3	305	410	294	234	443	429	162	95	110	20	322
21	QTA-20000 / SR3	330	478	295.5	255	519.5	435	166	97.5	110	20	350
22	QTA-22000 / SR3	330	478	295.5	255	519.5	435	166	97.5	145	20	350
23	QTA-25000 / SR3	330	478	295.5	255	519.5	435	166	97.5	145	20	413.5
24	QTA-31000 / SR3	346	462	307.5	270	546	442	180	101.5	145	20	420

■ FIG (3)



Sr. No.	MODELS	A	B	C	D	E	F	G	H	I	J	K
25	QTA-35000 / SR5	346	462	307.5	270	546	501	180	101.5	145	30	363
26	QTA-42000 / SR5	420	604	399	360	606	616	207	113	150	30	447
27	QTA-50000 / SR5	504	705	450	380	686.5	690	211.5	112.5	150	30	476

■ FIG (4)



Sr.No.	MODELS	A	B	C	D	E	F	G	H	I	J	K
28	QTA-65000 / SR4 / SR5	520	729	445	400	729	690	227	120	200	25	610
29	QTA-85000 / SR4 / SR5	520	729	445	400	729	700	227	120	200	25	625
30	QTA-0.1M / SR4 / SR5	520	729	445	400	729	700	227	120	200	25	625
31	QTA-0.15M / SR9 / SR10	750	999	614	540	890	864	311	150	250	40	834

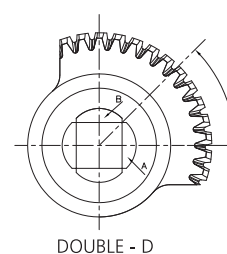
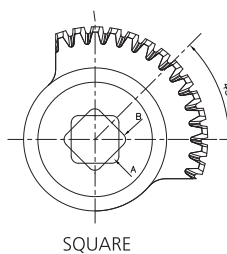
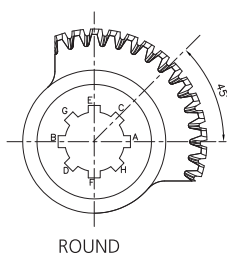
Manual Gear Box Selection Data

Sr. No.	Model	Output Torque		Mechanical Advantage +/-10 %	Gear Ratio	Turns to Close	ISO-5211 Valve Mounting	Max Stem Acceptance Dia (mm)	Approx. unit weight (Kg)	Recommended Hand Wheel Dia (mm)
		Nm	Ft-lbs							
1	QTA-150	150	110.6	8	36:1	9	F04, F05, F07	20	2.2	110
2	QTA-250	250	184.4	9.8	42:1	10.5	F05, F07	22	2.75	200
3	QTA-500	500	368.8	9.8	40:1	10	F07, F10	35	4.5	300
4	QTA-750	750	553.2	11.5	46:1	11.5	F07, F10, F12	35	5.6	400
5	QTA-1000	1000	737.6	12	46:1	11.5	F10, F12, F14	45	10	450
6	QTA-1250	1250	922	14	65:1	16.25	F10, F12, F14, F16	45	12	500
7	QTA-1500	1500	1106.3	16	62:1	15.5	F10, F12, F14, F16	50	16	500
8	QTA-2000	2000	1475.1	18	67:1	16.75	F10, F12, F14 F16, F20, F25	70	17	600
9	QTA-3000	3000	2212.7	18	62:1	15.5	F10, F12, F14, F16.	70	18	700
10	QTA-3750	3750	2765.9	22	80:1	20.0	F14, F16, F25	70	31.5	700
11	QTA-4000	4000	2950.2	22	87:1	21.75	F16, F25	85	40	700
12	QTA-4500	4500	3319	24	87:1	21.75	F16, F25	85	42	750
13	QTA-5500 / SR1	5500	4056.6	50	220.5:1	55	F16, F25	85	53	550
14	QTA-6000 / SR1	6000	4425.4	58	195.3:1	49	F16, F25	85	55	550
15	QTA-7000 / SR1	7000	5162.9	60	201.8:1	50.4	F16, F25, F30	95	68	600
16	QTA-8500 / SR2	8500	6269.3	73	286:1	71.5	F16, F25, F30	95	74	600
17	QTA-10000 / SR2	10000	7375.6	76	308:1	77	F25, F30	95	79	600
18	QTA-13000 / SR3	13000	9588.3	82	360:1	90	F25, F30	95	80	700
19	QTA-16000 / SR3	16000	11801	105	360:1	90	F25, F30	110	127	700
20	QTA-18500 / SR3	18500	13644.9	110	378:1	94.5	F25, F30	110	129	700
21	QTA-20000 / SR3	20000	14751.2	114.9	420:1	120	F25, F30, F35	110	137	750
22	QTA-22000 / SR3	22000	16226.4	114.9	420:1	120	F25, F30, F35	145	150	750
23	QTA-25000 / SR3	25000	18439.1	118.3	390:1	97.5	F25, F30, F35	145	153	800
24	QTA-31000 / SR3	31000	22864.4	121	403:1	100.3	F25, F30, F35	145	165	800
25	QTA-35000 / SR5	35000	25814.7	234.3	825:1	206.2	F25, F30, F35	145	185	800
26	QTA-42000 / SR5	42000	30977.6	266	1216:1	304	F25, F30, F35	150	292	850
27	QTA-50000 / SR5	50000	36878.1	289	1184:1	296	F30, F35, F40	150	386	850
28	QTA-65000 / SR5 / SR4	65000	47941.5	396	1743:1	436	F35, F40, F48	200	430	900
29	QTA-85000 / SR5/ SR4	85000	62692.8	590	2744:1	686	F35, F40, F48	200	442	900
30	QTA-0.1M / SR5/ SR4	100000	73756.2	623	2744:1	686	F35, F40, F48	200	463	900
31	QTA-0.15M / SR9 / SR10	150000	110634.3	810	3210:1	802.5	F40, F48	250	1050	1000

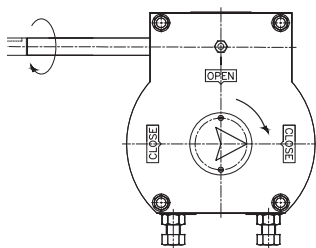
Material Specifications

Sr. No	Job Description	Material Description	Standards
1	Housing	Cast Iron-FG 220	IS 210 / ASTM A48 35B / 40B / GG-20
2	Cover	Cast Iron-FG 220	IS 210 / ASTM A48 35B / 40B / GG-20
3	Worm Wheel	SG Iron-500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
4	Input Shaft	EN8 / En19	45C8 / SAE 1040 / CK45 / C55 / SAE-4140 / 42CrMo4
5	Worm	En8 / 20MnCr5	45C8 / SAE 1040 / CK45 / 20MnCr1
6	Spur Housing	Cast Iron-FG 220	IS 210 / ASTM A48 35B / 40B / GG-20
7	Spur Cover	Cast Iron-FG 220	IS 210 / ASTM A48 35B / 40B / GG-20
8	Spur Pinion	EN19	C55 / SAE 4140 / 42CrMo4
9	Spur Gear	EN19	C55 / SAE 4140 / 42CrMo4
		SG Iron-500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
10	Bearing	Standard	----

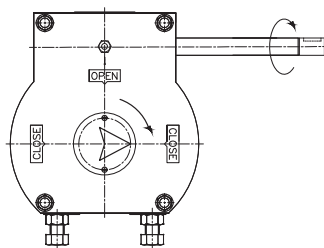
Worm Wheel Orientation Options as seen from Top View at Closed Position



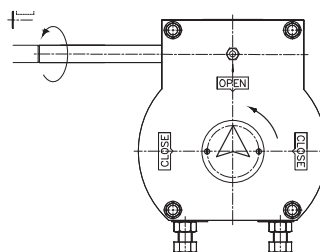
Input Shaft and Drive Orientation



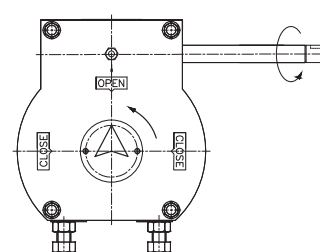
**Left Hand
Clockwise to
Close from Open**



**Right Hand
(Standard Supply)
Clockwise to
Close from Open**

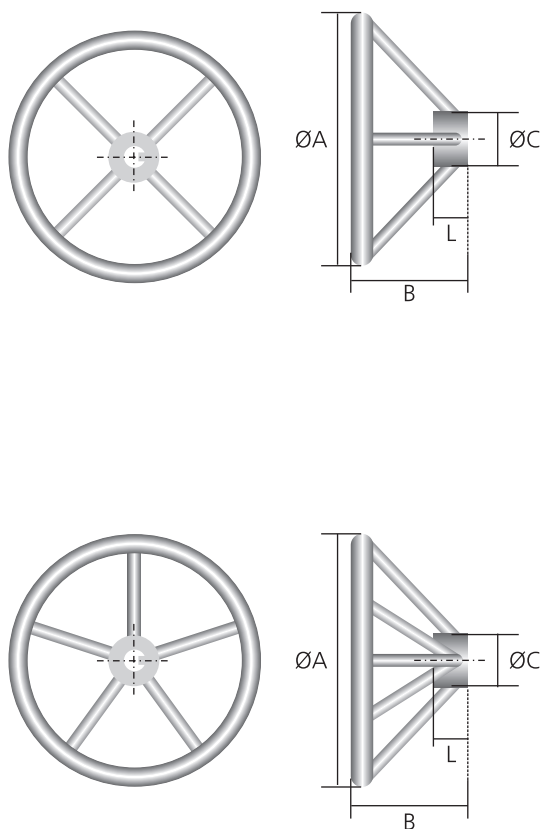


**Left Hand
Anti-Clockwise to
Open from Close**



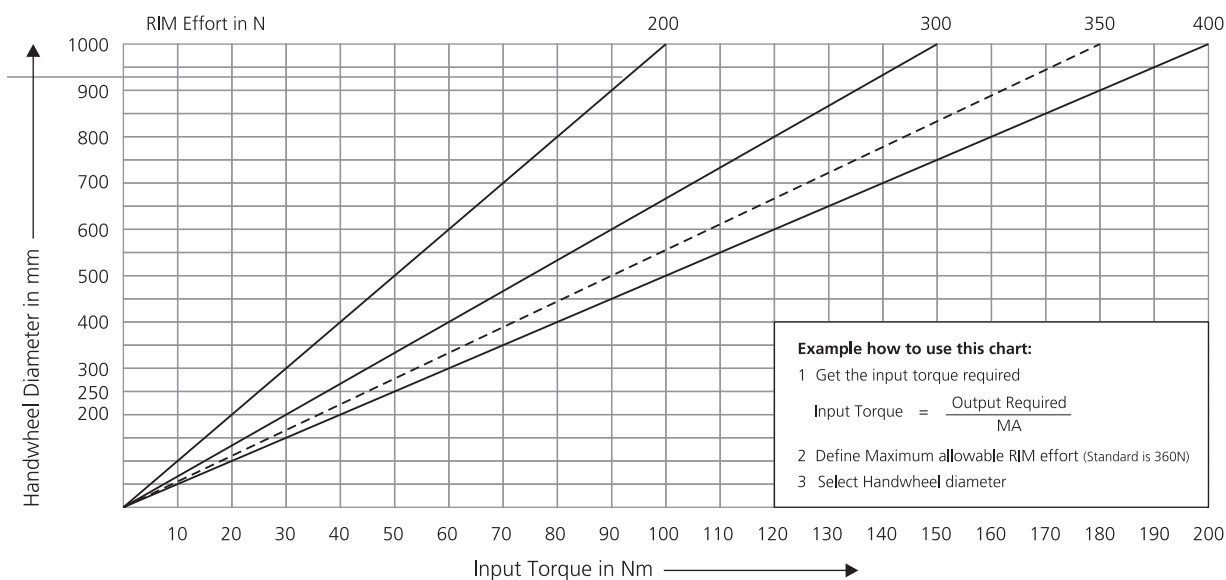
**Right Hand
Anti-Clockwise to
Open from Close**

Handwheel with Keyway



Sr. No	Handwheel Dia-A	Height B	Bush OD Dia-C	Bush Length-L	No. Of Spokes
1	110	52.5	25	20	3
2	200	80	25	25	4
		80	30	25	4
		80	40	30	4
3	250	80	40	30	4
4	300	100	40	25	4
		100	30	25	4
5	350	110	40	30	4
6	400	100	30	25	4
7	450	130	50	30	4
8	450	145	50	30	4
9	500	110	65	30	4
10	550	120	65	30	4
11	600	130	65	30	5
12	650	140	65	30	5
13	700	150	65	30	5
		150	65	50	5
14	750	160	65	30	5
15	800	160	65	30	5
16	850	160	65	30	5
		160	65	50	5
		160	65	50	5
17	900	160	65	30	5
18	1000	165	65	50	5

* Optional: Hand wheel with Dowel Hole



Ordering Information

QTA	Model			MOC-STD				VM		BORE		DECIMAL	ORIENTATION		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	0	4	-	A	A	A	-	E	X	S	D	0	A	X	-

Digit - 1,2,3		Digit -4	Digit - 5,6,7			Digit -8	Digit - 9,10		Digit - 11,12		Digit - 13		Digit - 14,15		Digit-16
Code	Model	Code	Code	MOC - Housing, Cover	Worm Wheel Worm Input Shaft	Code	Code	Mounting	Code	Bore	Code	Bore Digit	Code	Orientation	Code
A01	QTA-150	-	AAA	Cast Iron SG Iron EN8 AISI 1040 or EN8		-	BX	F04	***	Note 3	0	0	AX	A	-
A02	QTA-250		ACA	Cast Iron SG Iron 20MnCr5 AISI 1040 or EN8			CX	F05			1	.1	AB	AB	
A03	QTA-500		ACB	Cast Iron SG Iron 20MnCr5 AISI 4140 or EN19			DX	F07			2	.2	AC	AC	
A04	QTA-750		LAA	SG Iron SG Iron EN8 AISI 1040 or EN8			EX	F10			3	.3	AD	AD	
A05	QTA-1000		LCA	SG Iron SG Iron 20MnCr5 AISI 1040 or EN8			FX	F12			4	.4	AE	AE	
A06	QTA-1250		ACB	Cast Iron SG Iron 20MnCr5 AISI 4140 or EN19			GX	F14			5	.5	AF	AF	
A07	QTA-1500		PAM	CF8M SG Iron EN8 SS316			HX	F16			6	.6	AG	AG	
A08	QTA-2000(70)		PAA	CF8M SG Iron EN8 AISI 1040 or EN8			JX	F25			7	.7	AH	AH	
A09	QTA-2000(90)		PCM	CF8M SG Iron 20MnCr5 SS316			KX	F30			8	.8	BX	B	
A10	QTA-3000(70)		PMM	CF8M SG Iron SS316 SS316			LX	F35			9	.9	BC	BC	
A11	QTA-3000(90)		QAF	CF8 SG Iron EN8 SS304			MX	F40					BD	BD	
A12	QTA-3750		QCF	CF8 SG Iron 20MnCr5 SS304			NX	F48					BE	BE	
A13	QTA-4000(90)		LBB	SG Iron SG Iron EN19 AISI 4140 or EN19			UB	F04(UNC)					BF	BF	
A14	QTA-4000(115)		ABB	Cast Iron SG Iron EN19 AISI 4140 or EN19			UC	F05(UNC)					BG	BG	
A15	QTA-4500(90)		AAB	Cast Iron SG Iron EN8 AISI 4140 or EN19			UD	F07(UNC)					BH	BH	
A16	QTA-4500(115)		LAB	SG Iron SG Iron EN8 AISI 4140 or EN19			UE	F10(UNC)					CX	C	
A17	QTA-5500-SR1(90)		LCB	SG Iron SG Iron 20MnCr5 AISI 4140 or EN19			UF	F12(UNC)					CD	CD	
A18	QTA-5500-SR1(115)		WAA	WCB SG Iron EN8 AISI 1040 or EN8			UG	F14(UNC)					CE	CE	
A19	QTA-6000-SR1(90)		WCA	WCB SG Iron 20MnCr5 AISI 1040 or EN8			UH	F16(UNC)					CF	CF	
A20	QTA-6000-SR1(115)		WCB	WCB SG Iron 20MnCr5 AISI 4140 or EN19			UI	F25(UNC)					CG	CG	
A21	QTA-7000-SR2(90)		CAA	Cast Iron AB2 EN8 AISI 1040 or EN8			UJ	F30(UNC)					CH	CH	
A22	QTA-7000-SR2(125)		LBA	SG Iron SG Iron EN19 AISI 1040 or EN8			UK	F35(UNC)					DX	D	
A23	QTA-8500-SR2(90)		SBF	CF8 AB2 EN19 SS304			UL	F40(UNC)					DE	DE	
A24	QTA-8500-SR2(125)		QFF	CF8 SG Iron SS304 SS304			UM	F48(UNC)					DF	DF	
A25	QTA-10000-SR2		QBF	CF8 SG Iron EN19 SS304			***	Note 2					DG	DG	
A26	QTA-13000-SR3		UFF	CF8 CF8 SS304 SS304									DH	DH	
A27	QTA-16000-SR3		WBB	WCB SG Iron EN19 AISI 4140 or EN19									EX	E	
A28	QTA-18500-SR3		1BB	LCC SG Iron EN19 AISI 4140 or EN19									EF	EF	
A29	QTA-20000-SR3		***	Note 1									EG	EG	
A30	QTA-22000-SR3												EH	EH	
A31	QTA-25000-SR3												FX	F	
A32	QTA-31000-SR3A												FG	FG	
A33	QTA-35000-SR5A												FH	FH	
A34	QTA-42000-SR5												GX	G	
A35	QTA-50000-SR5												GH	GH	
A36	QTA-65000-SR5-SR4												HX	H	
A37	QTA-85000-SR5-SR4												A1	ABE	
A38	QTA-0.1M-SR5-SR4												A2	ABF	
A39	QTA-0.15M-SR9-SR10												C1	CDG	
													C2	CDH	
													E1	EFA	
													E2	EFB	
													G1	GHC	
													G2	GHD	
													YY	ABEF	
													ZZ	CDGH	
													XX	NA	

Note 1	For more MOC combinations, please contact Q-Tork
Note 2	For multiple Gearbox mounting PCD, please contact Q-Tork
Note 3	Detailed coding is mentioned on Page 62 of this Catalogue
Note 4	For additional type and size, Please contact Q-Tork
Note 5	For additional accessories, please specify
Note 6	For more specials, please specify

HAND WHEEL		ACCESSORIES		IP / LA		SPECIAL		
17	18	19	20	21	22	23	24	25
K	F	X	X	1	-	X	X	X

Digit - 17,18		Digit - 19,20		Digit - 21		Digit-22	Digit - 23,24,25	
Code	Handwheel / Chain Wheel	Code	Accessories	Code	IP rating / Padlock	Code	Code	Special
KA	MS Fabricated H/W:- Dia 110mm Key Type	BF	Bottom side Bracket & Adaptor	1	Enclosure type:- IP65 Without Pad Lock	-	AAA	NCP shaft & Hardware SS304
KB	MS Fabricated H/W:- Dia 200mm Key Type	BG	bottom side Adaptor	2	Enclosure type:- IP67 Without Pad Lock		AAH	Hardware SS316, Gear box RAL5015
KC	MS Fabricated H/W:- Dia 250mm Key Type	SA	Shaft Extension	3	Enclosure type:- IP68 Without Pad Lock		AAL	Gear box and handwheel RAL3001
KD	MS Fabricated H/W:- Dia 300mm Key Type	SB	Chain	4	Enclosure type:- Submersible Without Pad Lock		AAQ	Dowel hole arrangement
KE	MS Fabricated H/W:- Dia 350mm Key Type	TA	With Limit Switch, Bracket, Adapter	5	Enclosure type:- IP65 With Pad Lock		AAS	Gear Box RAL 9005
KF	MS Fabricated H/W:- Dia 400mm Key Type	TB	Limit Switch Arrangement	6	Enclosure type:- IP67 With Pad Lock		AAU	Hardware SS316, Gear box RAL9005
KG	MS Fabricated H/W:- Dia 450mm Key Type	TD	Position Indicator Scale	7	Enclosure type:- IP68 With Pad Lock		AAY	Gear box RAL5015
KH	MS Fabricated H/W:- Dia 500mm Key Type	XX	No accessories	8	Enclosure type:- Submersible With Pad Lock		ABE	Hardware SS304
KJ	MS Fabricated H/W:- Dia 550mm Key Type	**	Note 5				ABF	Hardware SS316
KK	MS Fabricated H/W:- Dia 600mm Key Type						ABJ	Gear box unpainted
KL	MS Fabricated H/W:- Dia 630mm Key Type						ABQ	Both Side Shaft
KM	MS Fabricated H/W:- Dia 650mm Key Type						ABS	Solid Hand wheel
KN	MS Fabricated H/W:- Dia 700mm Key Type						ACJ	Low Temperature Grease
KP	MS Fabricated H/W:- Dia 710mm Key Type						XXX	without any specials
KQ	MS Fabricated H/W:- Dia 750mm Key Type						***	Note 6
KR	MS Fabricated H/W:- Dia 800mm Key Type							
KS	MS Fabricated H/W:- Dia 850mm Key Type							
KT	MS Fabricated H/W:- Dia 900mm Key Type							
KU	MS Fabricated H/W:- Dia 1000mm Key Type							
KV	MS Fabricated H/W:- Dia 175mm Key Type							
WA	Chain Wheel C.I:- Dia 150mm							
WB	Chain Wheel C.I:- Dia 220mm							
WC	Chain Wheel C.I:- Dia 300mm							
WD	Chain Wheel C.I:- Dia 380mm							
WE	Chain Wheel C.I:- Dia 400mm							
WF	Chain Wheel C.I:- Dia 500mm							
WG	Chain Wheel C.I:- Dia 585mm							
YA	Chain Wheel D.I:- Dia 150mm							
YB	Chain Wheel D.I:- Dia 220mm							
YC	Chain Wheel D.I:- Dia 300mm							
YD	Chain Wheel D.I:- Dia 380mm							
YE	Chain Wheel D.I:- Dia 400mm							
YF	Chain Wheel D.I:- Dia 500mm							
YG	Chain Wheel D.I:- Dia 585mm							
XX	Without Handwheel							
**	Note 4							

Build Your Code :

Blank Your Code :

QTA	MODEL			MOC-STD.				VM		BORE		DECIMAL	ORIENTATION			HAND WHEEL		ACCESSORIES		IP / LA		SPECIAL			
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	

Actual Product Photos - Optional Accessories

Locking arrangement



Through Shaft arrangement



Chain Wheel arrangement



Limit Switch Box (LSB) arrangement



Stainless Steel Gear Box and Hand wheel



* Optional: Chain wheel arrangement with 10° variable for 360° adjustment



612 Series | Manual Override Gear Operators

Q-Tork 612 series Manual Override (MOR) are designed and used for manually overriding pneumatic actuators mainly for industrial application

The compact robust designed product has optimum ratio to reduce efforts during its operation

Application

612 series MOR are widely used for its reliability where their are double acting and spring return pneumatic actuators

All MOR are mounted between the valve and actuator

Features & Benefits

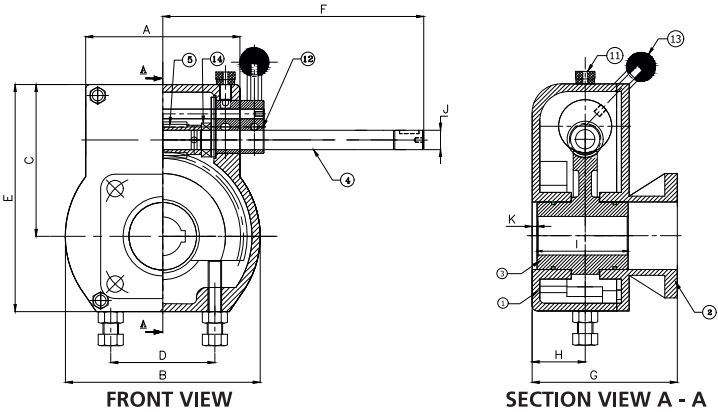
- Graded CI housing and cover
- High performance axial bearing
- Light weight with rugged design
- Maintenance free
- Temperature range -30°C to +120°C
- Epoxy primer coated housing and cover
- Environmental friendly high performance grease
- Alloy steel-worm & input shaft
- Powder coated handwheel

Optional

- Offering MOC:
 - ▶ Housing/ Cover- Carbon steel, Ductile Iron, Stainless Steel
 - ▶ Quadrant - Al. Bronze
- Submersibility to IP-67
- Stainless steel - shaft, handwheel and hardware
- Lubricants in temperature variants can be provided.

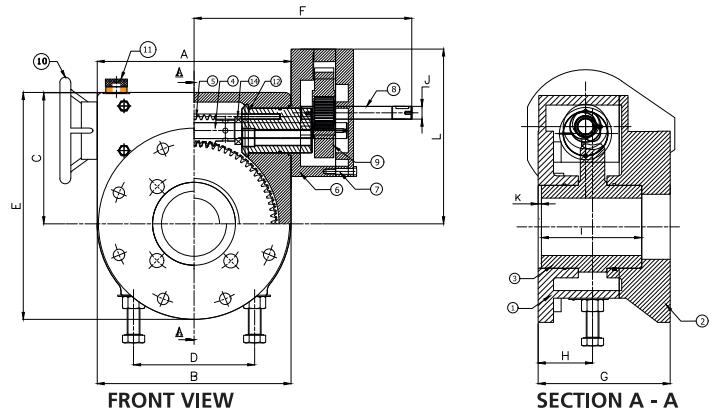
■ Dimensional Specifications

■ FIG (1)



Sr. No.	MODELS	A	B	C	D	E	F	G	H	I	J	K
1	QTM-100	89	72	66	40	93	119	86.5	32	47.5	10	2
2	QTM-250	106	98	81	44	121	215	86	28.5	50	12	2
3	QTM-450	115	132	97.5	70	142.5	165	123	41	73	15	2
4	QTM-750	117	125	109	70	165	185.5	123	45	72.5	17	2
5	QTM-1000	154	180	120.5	95	185.5	190	148.5	59	112.5	20	4
6	QTM-1500	160	202	157.5	108	235.5	230	150.5	55	94.5	20	5
7	QTM-2500	204	218	165	108	250	237	171	66	122	20	6
8	QTM-3500	185	296	203	150	351	265	195	78	128	20	7

■ FIG (2)



Sr. No.	MODELS	A	B	C	D	E	F	G	H	I	J	K	L
9	QTM-5000 / MR1	270	300	193	180	343	284.5	204	84	156	20	5	234
10	QTM-7500 / MR2	304	326	235	190	385	327	216	84	160	20	4	297
11	QTM-10000 / MR2	362	367	367	210	440	370	242	98	198.5	20	6	362.5
12	QTM-15000 / MR3	400	403	282	228	483.5	389.5	243	108	211	20	6.5	400
13	QTM-35000 / MR4	460	510	360	300	567	500	330	119	292	20	8	429

Manual Overrides Gear Box Selection Data

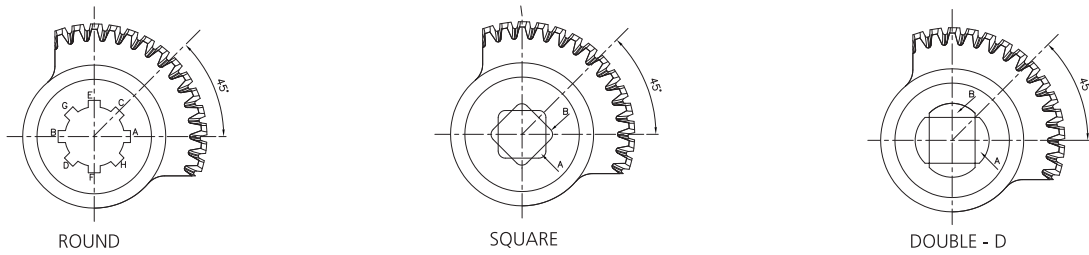
Sr. No	QTM Model	Output Torque		Mechanical Advantages +/-10%	Gear Ratio	Turns To Close	ISO 5211 Valve Mounting	ISO 5211 Actuator Mounting	Max Stem Acceptance Dia (mm)	Approx. Unit Weight (Kgs)	Recommended Hand Wheel Dia. (mm)
		Nm	Ft-Lbf								
1	QTM-100*	100	73.8	8	36:1	9	F04, F05	F04	16	3.2	110
								F05, F07	20	3.7	
2	QTM-250	250	184.4	9	38:1	9.5	F05, F07	F05	18	4.2	200
								F07, F10	22	6	
3	QTM-450	450	331.9	10	43:1	10.75	F07, F10	F07, F10	30	8	300
4	QTM-750	750	553.2	11.5	46:1	11.5	F07, F10, F12	F07	33	9.7	400
								F10, F12	40	10.2	
5	QTM-1000	1000	737.6	12.5	48:1	12	F10, F12, F14	F10, F12, F14	50	15	500
6	QTM-1500	1500	1106.3	18	67:1	16.75	F10, F12, F14, F16	F10, F12, F14	50	22	500
7	QTM-2500	2500	1843.9	17.5	60:1	15	F12, F14, F16	F12, F14, F16	65	39	850
8	QTM-3500	3500	2581.5	23.5	80:1	20	F14, F16, F25	F16, F25	80	58	850
9	QTM-5000/MR1	5000	3687.8	68.9	231:1	57.75	F16, F25	F16, F25	90	65	500
10	QTM-7500/MR2	7500	5531.7	74	340:1	85	F16, F25	F16	90	115	600
								F25	100	118	
11	QTM-10000/MR2	10000	7375.6	85	390:1	97.5	F25, F30	F25, F30	120	130	700
12	QTM-15000/MR3	15000	11063.4	94	408:1	102	F25, F30	F25, F30, F35	140	160	750
13	QTM-35000/MR4	35000	25814.7	280	1288:1	322	F30, F35	F30, F35	160	285	750

* QTM-100 Standard Handwheel is with Dowel Hole

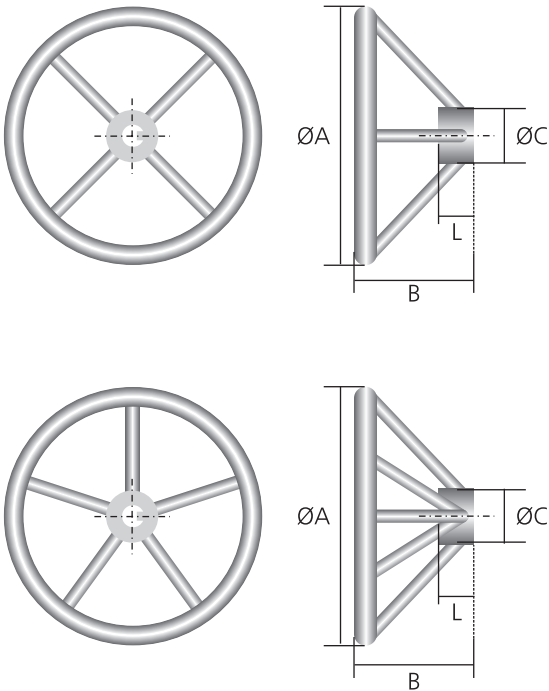
Material Specifications

Sr. No	Job Description	Material Description	Standards
1	Housing	Cast Iron-FG 220	IS 210 / ASTM A48 35B/40B/GG-20
2	Cover	Cast Iron-FG 220	IS 210 / ASTM A48 35B/40B/GG-20
3	Worm Wheel	SG Iron-500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
4	Worm	EN8 / 20MnCr5	45C8 / SAE 1040 / CK45
5	Input Shaft	EN8 / EN19	45C8 / SAE 1040 / CK45 / C55 / SAE-4140 / 42CrMo4
6	Thrust Bearing	Standard	----
7	Eccentric Bush	Mild Steel	----
8	Spur Housing	Cast Iron-FG 220	IS 210 / ASTM A48 35B/40B/GG-20
9	Spur Cover	Cast Iron-FG 220	IS 210 / ASTM A48 35B/40B/GG-20
10	Spur Gear	EN19	C55 / SAE 4140 / 42CrMo4
11	Spur Pinion	EN19	C55 / SAE 4140 / 42CrMo4
12	Selector assembly	Mild Steel	----
13	Lever	Mild Steel	----
14	Clutch & Declutch hand wheel	Mild Steel	----

Worm Wheel Orientation Options as seen from Top View at Closed Position

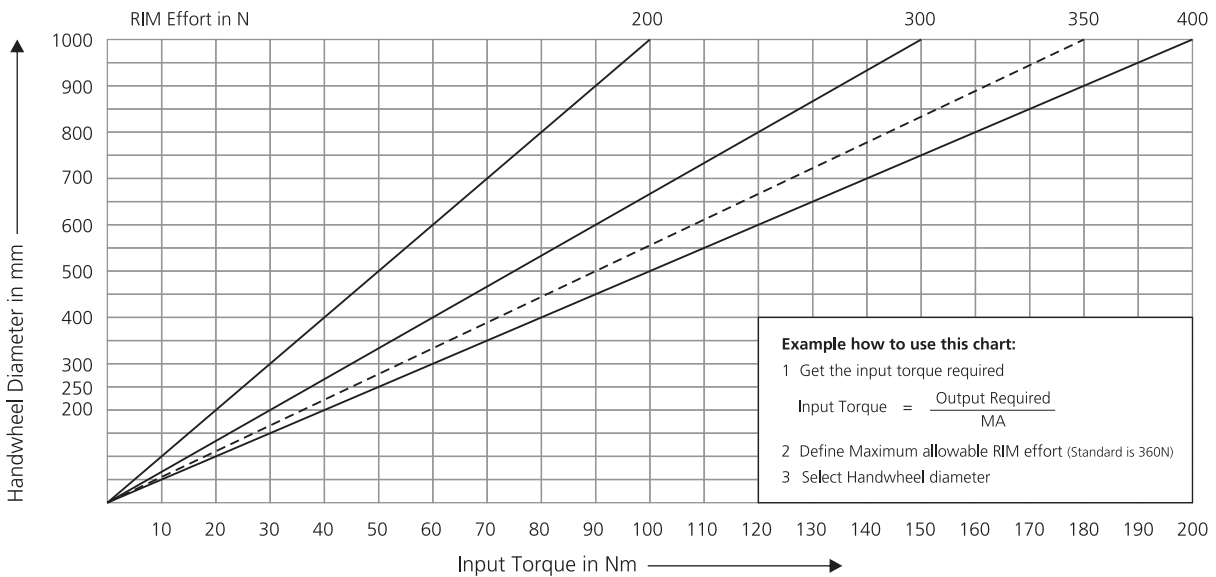


Handwheel Specification: For Keyway

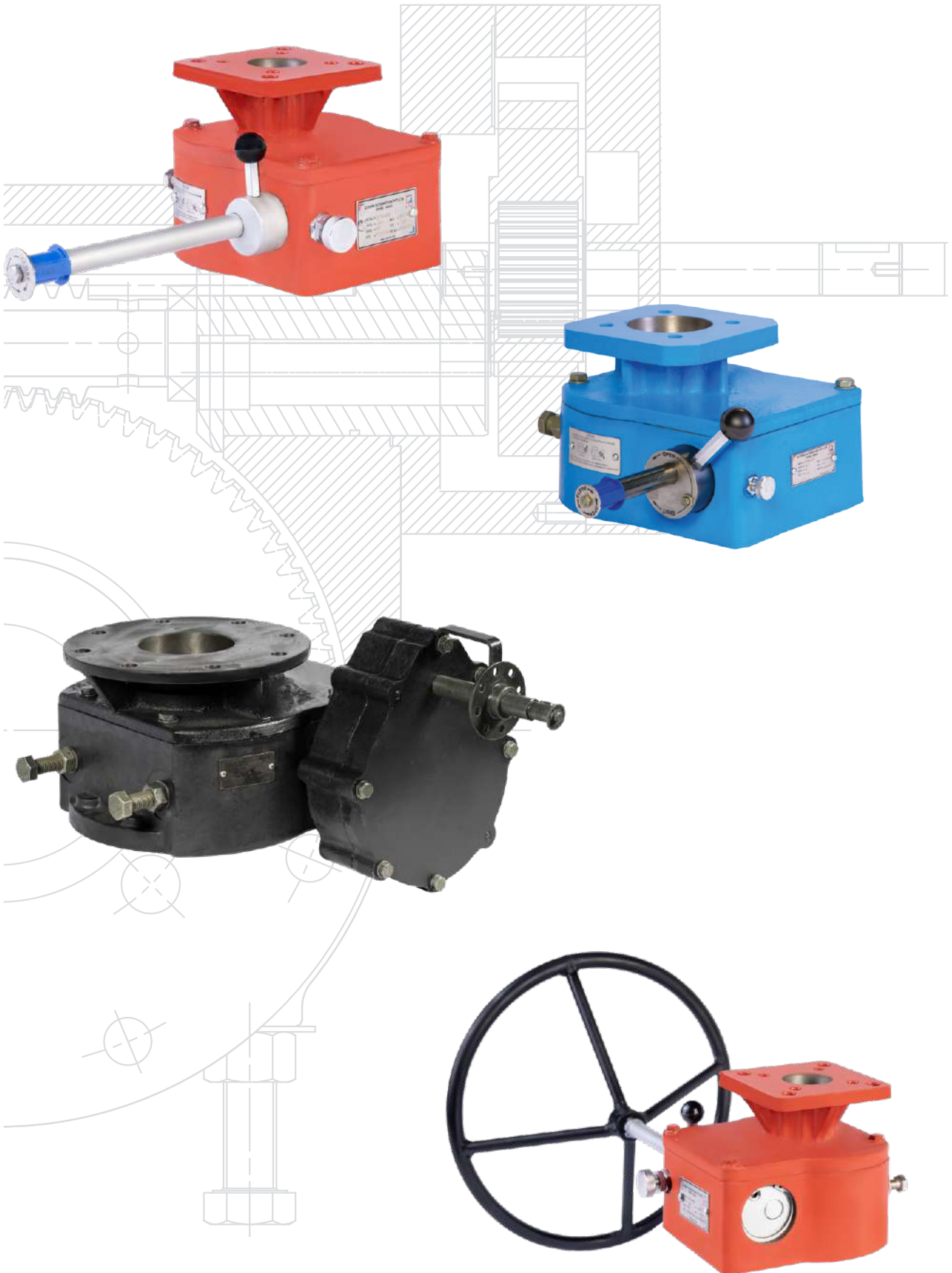


Sr. No	Handwheel Dia-A	Height B	Bush OD Dia-C	Bush Length-L	No. Of Spokes
1	70	30	20	30	3
2	200	80	25	25	4
		50	35	35	3
3	250	80	40	30	4
			30	25	
4	300	100	40	30 / 25	4
		75	45	50	3
5	350	110	40	30	4
		100	30	25	
6	400	130	50	30	4
7	450	145	50	30	4
8	500	110	65	30	4
9	550	120	65	30	4
10	600	130	65	30	5
11	650	140	65	30	5
12	700	150	65	30	5
13	750	160	65	30	5
				50	5
14	850	160	65	30	5
15	900	160	65	30	5
16	1000	165	65	50	5

* Optional: Hand wheel with Dowel Hole



Actual Product Photos



Ordering Information

QTM	Model			MOC-STD				VM & AM		BORE		DECIMAL
1	2	3	4	5	6	7	8	9	10	11	12	13
M	0	5	-	A	A	A	-	E	G	B	T	0

Digit - 1,2,3		Digit -4	Digit - 5,6,7				Digit - 8		Digit - 9,10		Digit - 11,12		Digit - 13	
Code	Model	-	Code	MOC - Housing, Cover Worm Wheel Worm Input Shaft			Code	Code	Mounting		Code	Bore	Code	Bore Digit
M01	QTM-100		AAA	Cast Iron SG Iron EN8 AISI 1040 or EN8			-	BB	Valve Mounting-F04, Actuator Mounting-F04		Note 3		0	0
M02	QTM-250		ACA	Cast Iron SG Iron 20MnCr5 AISI 1040 or EN8				BC	Valve Mounting-F04, Actuator Mounting-F05				1	.1
M03	QTM-450		ACB	Cast Iron SG Iron 20MnCr5 AISI 4140 or EN19				CB	Valve Mounting-F05, Actuator Mounting-F04				2	.2
M04	QTM-750		LAA	SG Iron SG Iron EN8 AISI 1040 or EN8				CC	Valve Mounting-F05, Actuator Mounting-F05				3	.3
M05	QTM-1000		LCA	SG Iron SG Iron 20MnCr5 AISI 1040 or EN8				CD	Valve Mounting-F05, Actuator Mounting-F07				4	.4
M06	QTM-1500		ACB	Cast Iron SG Iron 20MnCr5 AISI 4140 or EN19				DC	Valve Mounting-F07, Actuator Mounting-F05				5	.5
M07	QTM-2500		PAM	CF8M SG Iron EN8 SS316				DD	Valve Mounting-F07, Actuator Mounting-F07				6	.6
M08	QTM-3500		PAA	CF8M SG Iron EN8 AISI 1040 or EN8				DE	Valve Mounting-F07, Actuator Mounting-F10				7	.7
M09	QTM-5000-MR1		PCM	CF8M SG Iron 20MnCr5 SS316				DF	Valve Mounting-F07, Actuator Mounting-F12				8	.8
M10	QTM-7500-MR2		PMM	CF8M SG Iron SS316 SS316				ED	Valve Mounting-F10, Actuator Mounting-F07				9	.9
M11	QTM-10000-MR2		QAF	CF8 SG Iron EN8 SS304				EE	Valve Mounting-F10, Actuator Mounting-F10					
M12	QTM-15000-MR3		QCF	CF8 SG Iron 20MnCr5 SS304				EF	Valve Mounting-F10, Actuator Mounting-F12					
M14	QTM-35000-MR4		LBB	SG Iron SG Iron EN19 AISI 4140 or EN19				EG	Valve Mounting-F10, Actuator Mounting-F14					
			ABB	Cast Iron SG Iron EN19 AISI 4140 or EN19				FD	Valve Mounting-F12, Actuator Mounting-F07					
		AAB	Cast Iron SG Iron EN8 AISI 4140 or EN19				FE	Valve Mounting-F12, Actuator Mounting-F10						
		LAB	SG Iron SG Iron EN8 AISI 4140 or EN19				FF	Valve Mounting-F12, Actuator Mounting-F12						
		LCB	SG Iron SG Iron 20MnCr5 AISI 4140 or EN19				FG	Valve Mounting-F12, Actuator Mounting-F14						
		WAA	WCB SG Iron EN8 AISI 1040 or EN8				FH	Valve Mounting-F12, Actuator Mounting-F16						
		WCA	WCB SG Iron 20MnCr5 AISI 1040 or EN8				GE	Valve Mounting-F14, Actuator Mounting-F10						
		WCB	WCB SG Iron 20MnCr5 AISI 4140 or EN19				GF	Valve Mounting-F14, Actuator Mounting-F12						
		CAA	Cast Iron AB2 EN8 AISI 1040 or EN8				GG	Valve Mounting-F14, Actuator Mounting-F14						
		LBA	SG Iron SG Iron EN19 AISI 1040 or EN8				GH	Valve Mounting-F14, Actuator Mounting-F16						
		SBF	CF8 AB2 EN19 SS304				GJ	Valve Mounting-F14, Actuator Mounting-F25						
		QFF	CF8 SG Iron SS304 SS304				HE	Valve Mounting-F16, Actuator Mounting-F10						
		QBF	CF8 SG Iron EN19 SS304				HF	Valve Mounting-F16, Actuator Mounting-F12						
		UFF	CF8 CF8 SS304 SS304				HG	Valve Mounting-F16, Actuator Mounting-F14						
		WBB	WCB SG Iron EN19 AISI 4140 or EN19				HH	Valve Mounting-F16, Actuator Mounting-F16						
		1BB	LCC SG Iron EN19 AISI 4140 or EN19				HJ	Valve Mounting-F16, Actuator Mounting-F25						
		***	Note 1				JE	Valve Mounting-F25, Actuator Mounting-F10						
							JG	Valve Mounting-F25, Actuator Mounting-F14						
							JH	Valve Mounting-F25, Actuator Mounting-F16						
							JJ	Valve Mounting-F25, Actuator Mounting-F25						
							JK	Valve Mounting-F25, Actuator Mounting-F30						
							JL	Valve Mounting-F25, Actuator Mounting-F35						
							KE	Valve Mounting-F30, Actuator Mounting-F10						
							KG	Valve Mounting-F30, Actuator Mounting-F14						
							KH	Valve Mounting-F30, Actuator Mounting-F16						
							KJ	Valve Mounting-F30, Actuator Mounting-F25						
							KK	Valve Mounting-F30, Actuator Mounting-F30						
							KL	Valve Mounting-F30, Actuator Mounting-F35						
							LE	Valve Mounting-F35, Actuator Mounting-F10						
							LG	Valve Mounting-F35, Actuator Mounting-F14						
							LH	Valve Mounting-F35, Actuator Mounting-F16						
							LJ	Valve Mounting-F35, Actuator Mounting-F25						
							LK	Valve Mounting-F35, Actuator Mounting-F30						
							LL	Valve Mounting-F35, Actuator Mounting-F35						
							**	Note 2						

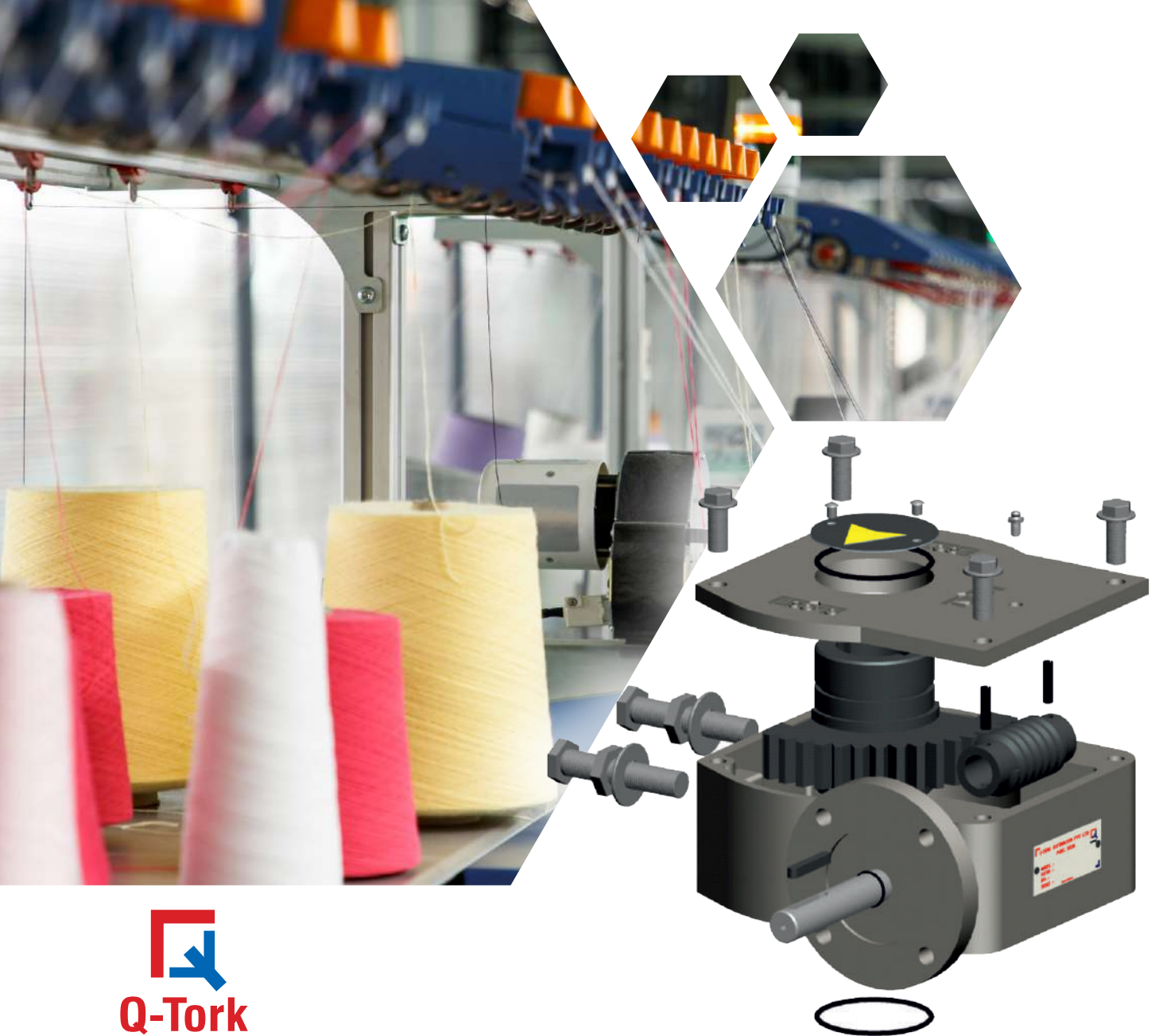
Note 1	For more MOC combinations, please contact Q-Tork
Note 2	For multiple Gearbox mounting PCD, please contact Q-Tork
Note 3	Detailed coding is mentioned on Page 62 of this Catalogue
Note 4	For additional accessories, please specify
Note 5	For more specials, please specify

ORIENTATION			HAND WHEEL			ACCESSORIES		IP / LA			SPECIAL		
14	15	16	17	18	19	20	21	22	23	24	25		
A	B	-	K	H	X	X	1	-	X	X	X		

Digit - 14,15		Digit - 16	Digit - 17,18			Digit - 19,20		Digit - 21			Digit-22	Digit - 23,24,25	
Code	Orientation	Code	Code	Handwheel / Chain Wheel	Code	Accessories	Code	IP rating / Padlock	Code	Code	Special		
AX	A	-	KA	MS Fabricated H/W:- Dia 110mm Dowel Hole	XX	No accessories	1	Enclosure type:- IP65 Without Pad Lock	-	AAA	NCP shaft & Hardware SS304		
AB	AB		KB	MS Fabricated H/W:- Dia 200mm Key Type	**	Note 5	2	Enclosure type:- IP67 Without Pad Lock		AAH	Hardware SS316, Gear box RAL5015		
AC	AC		KC	MS Fabricated H/W:- Dia 250mm Key Type			3	Enclosure type:- IP68 Without Pad Lock		AAL	Gear box and handwheel RAL3001		
AD	AD		KD	MS Fabricated H/W:- Dia 300mm Key Type			4	Enclosure type:- Submersible Without Pad Lock		AAS	Gear Box RAL 9005		
AE	AE		KE	MS Fabricated H/W:- Dia 350mm Key Type			5	Enclosure type:- IP65 With Pad Lock		AAU	Hardware SS316, Gear box RAL9005		
AF	AF		KF	MS Fabricated H/W:- Dia 400mm Key Type			6	Enclosure type:- IP67 With Pad Lock		AAY	Gear box RAL5015		
AG	AG		KG	MS Fabricated H/W:- Dia 450mm Key Type			7	Enclosure type:- IP68 With Pad Lock		ABE	Hardware SS304		
AH	AH		KH	MS Fabricated H/W:- Dia 500mm Key Type			8	Enclosure type:- Submersible With Pad Lock		ABF	Hardware SS316		
BX	B		KJ	MS Fabricated H/W:- Dia 550mm Key Type						ABJ	Gear box unpainted		
BC	BC		KK	MS Fabricated H/W:- Dia 600mm Key Type						ABS	Solid Hand wheel		
BD	BD		KL	MS Fabricated H/W:- DIA 630mm Key Type						ACJ	Low Temperature Grease		
BE	BE		KM	MS Fabricated H/W:- Dia 650mm Key Type						XXX	without any specials		
BF	BF		KN	MS Fabricated H/W:- Dia 700mm Key Type						***	Note 6		
BG	BG		KP	MS Fabricated H/W:- Dia 710mm Key Type									
BH	BH		KQ	MS Fabricated H/W:- Dia 750mm Key Type									
CX	C		KR	MS Fabricated H/W:- Dia 800mm Key Type									
CD	CD		KS	MS Fabricated H/W:- Dia 850mm Key Type									
CE	CE		KT	MS Fabricated H/W:- Dia 900mm Key Type									
CF	CF		KU	MS Fabricated H/W:- Dia 1000mm Key Type									
CG	CG		KV	MS Fabricated H/W:- Dia 175mm Key Type									
CH	CH		XX	Without Handwheel									
DX	D		**	Note 4									
DE	DE												
DF	DF												
DG	DG												
DH	DH												
EX	E												
EF	EF												
EG	EG												
EH	EH												
FX	F												
FG	FG												
FH	FH												
GX	G												
GH	GH												
HX	H												
A1	ABE												
A2	ABF												
C1	CDG												
C2	CDH												
E1	EFA												
E2	EFB												
G1	GHC												
G2	GHD												
YY	ABEF												
ZZ	CDGH												
XX	NA												

Build Your Code :

Grand Total Code 1																										
QTM	MODEL			MOC-STD.				VM			BORE		DECIMAL	ORIENTATION			HAND WHEEL		ACCESSORIES			IP / LA		SPECIAL		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					



143 Series | Quarter-turn Worm Gear Operators for Electrical Actuation

Q-Tork 143 series Electrical Quarter turn Worm Gear Operators are designed for motorized application.

Application

143 series Electrical quarter turn worm gear boxes are used in operation of Ball valve, Butterfly valve, Plug valve and also for Dampers.

Features & Benefits

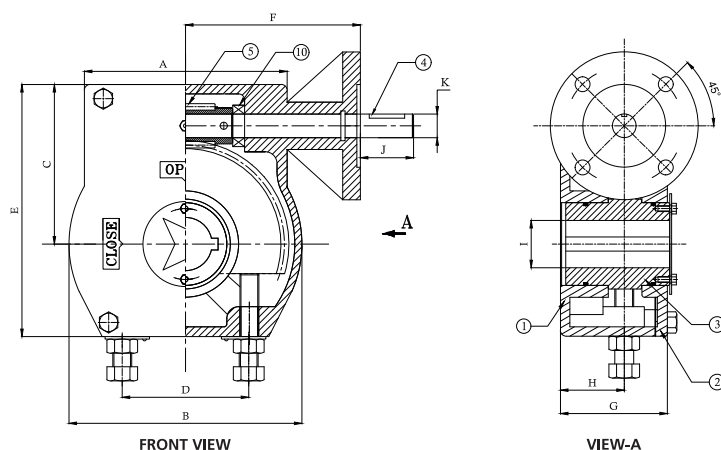
- Graded CI - Housing and Cover
- Alloy Steel - Worm and input Shaft
- All critical components are heat treated to ensure longer serviceable life
- Comprehensive gear ratios with spur reduction unit
- Stroke – 0 to 90° ($\pm 5^\circ$ adjustable)
- High performance axial thrust bearing
- Temperature range -30°C to $+120^\circ\text{C}$
- Light weight with rugged design
- Environmental friendly high performance grease
- Maintenance Free

Optional

- Offering MOC:
 - ▶ Housing/ Cover- Carbon steel, Ductile Iron, Stainless Steel
 - ▶ Quadrant - Al. Bronze
- Stainless steel - Input shaft
- Submersibility to IP67
- Lubricants in temperature variants can be provided.

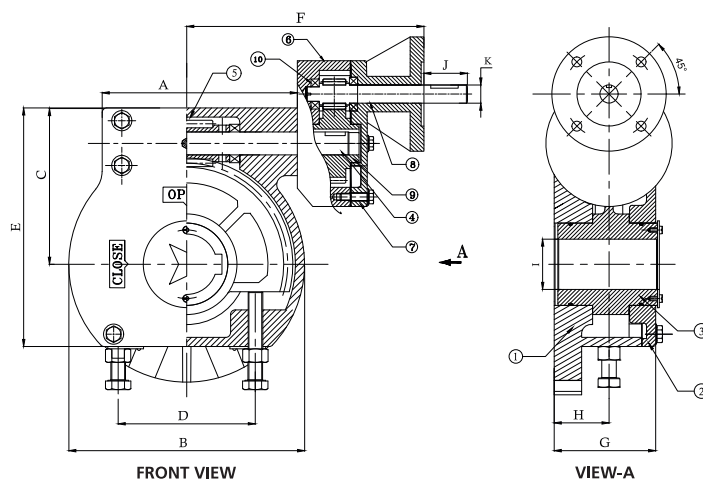
Dimensional Specifications

FIG (1)



Sr. No.	MODELS	A	B	C	D	E	F	G	H	I	J	K
1	QTE-700	99	116	91	68	134	131	63.5	36	35	45	20
2	QTE-2000	172	197	135	108	213	148	90.5	54	50	45	20
3	QTE-3000	185	296	171	150	261	220	109	62	70	60	25
4	QTE-3600	213	300	195	150	305	199.5	118.5	70.5	85	60	25

FIG (2)



Sr. No.	MODELS	A	B	C	D	E	F	G	H	I	J	K
5	QTE-4000/ER1	213.5	258	171	146	261	259.5	111	60	85	47.5	20
6	QTE-6000/ER2	253.5	350	202	146	312	284	146.5	91.5	95	49.5	20
7	QTE-7500/ER2	257	304	220	170	329	264.5	146.5	89.5	95	49.5	20
8	QTE-10000/ER3	280	353	239	185	415	400	156	87	95	50	25
9	QTE-15000/ER3	305	410	267.5	234	443	419	162	95	110	50	25
10	QTE-20000/ER3	330	478	295.5	255	519.5	425	166	97.5	145	50	25
11	QTE-30000/ER5	346	462	307.5	270	546	475	180	101.5	145	50	30
12	QTE-45000/ER5	504	706	450	380	686.5	622	211.5	112.5	150	50	30

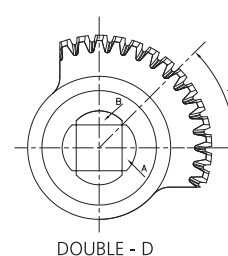
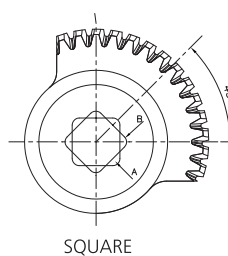
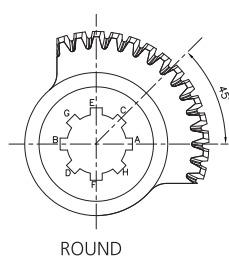
Electrical Gear Box Selection Data

Sr. No.	Model	Output Torque		Mechanical Advantage +/- 10 %	Gear Ratio	Turns to Close	ISO 5211 Valve Mounting	ISO 5211 Actuator Mounting	Max Stem Acceptance Dia (mm)	Approx. unit weight (Kg)
		Nm	Ft-Lbf							
1	QTE-700	700	516	11.5	46:1	11.5	F07, F10, F12	F10	35	7.5
2	QTE-2000	2000	1475	18	67:1	16.75	F10, F12, F14, F16	F10	50	21
3	QTE-3000	3000	2212.5	20.4	74:1	18.5	F14, F16, F25	F10	70	35
4	QTE-3600	3600	2655	22	87:1	21.75	F16, F25	F10	85	40
5	QTE-4000/ER1	4000	2950	44	177:1	44.25	F12, F14, F16, F25	F10	85	44
6	QTE-6000/ER2	6000	4425	60	256:1	64	F16, F25, F30	F10	95	73
7	QTE-7500/ER2	7500	5531	73	260:1	65	F16, F25, F30	F10	95	77
8	QTE-10000/ER3	10000	7375	82	360:1	90	F25, F30	F10, F14	95	114
9	QTE-15000/ER3	15000	11062.5	110	378:1	94.5	F25, F30	F10, F14	110	134
10	QTE-20000/ER3	20000	14750	118.3	390:1	97.5	F25, F30, F35	F10, F14	145	179
11	QTE-30000/ER5	30000	22125	234.3	825:1	206.25	F25, F30, F35	F14	145	296
12	QTE-45000/ER5	45000	33187.5	289	1184:1	296	F30, F35, F40	F14	150	393

Material Specifications

Sr. No	Job Description	Material Description	Standards
1	Housing	Cast Iron-FG 220	IS 210 / ASTM A48 35B / 40B / GG-20
2	Cover	Cast Iron-FG 220	IS 210 / ASTM A48 35B / 40B / GG-20
3	Worm Wheel	SG Iron-500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
4	Input Shaft	En19	45C8 / SAE 1040 / CK45 / C55 / SAE-4140 / 42CrMo4
5	Worm	20MnCr5	45C8 / SAE 1040 / CK45 / 20MnCr1
6	Spur Housing	Cast Iron-FG 220	IS 210 / ASTM A48 35B / 40B / GG-20
7	Spur Cover	Cast Iron-FG 220	IS 210 / ASTM A48 35B / 40B / GG-20
8	Spur Pinion	EN19	C55 / SAE 4140 / 42CrMo4
9	Spur Gear	EN19	C55 / SAE 4140 / 42CrMo4
		SG Iron-500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
10	Bearing	Standard	----

Worm Wheel Orientation Options as seen from Top View at Closed Position



The image displays three industrial gearboxes from Q-TORK Automation Pvt. Ltd. Two are red and one is blue. Each unit features a top-mounted control panel with a yellow triangular indicator and a black adjustment knob. The red gearboxes have two large horizontal mounting flanges on the side, while the blue one has a single large flange and a smaller blue output shaft. A technical drawing of a gearbox assembly is visible in the background. A label on the blue gearbox provides the following specifications: Q-TORK AUTOMATION PVT. LTD., PUNE, INDIA; MODEL: Q-TORK; RATIO: 10; GEAR: 1; SP. NO: 1; and www.q-tork.com.

Ordering Information

QTE	Model			MOC-STD				VM & AM		BORE		DECIMAL
1	2	3	4	5	6	7	8	9	10	11	12	13
E	0	3	-	A	C	A	-	H	E	C	Q	0

Digit - 1,2,3		Digit -4	Digit - 5,6,7				Digit -8	Digit - 9,10		Digit - 11,12		Digit - 13			
Code	Model	-	AAA	MOC - Housing, Cover Worm Wheel Worm Input Shaft				Code	Code	Mounting		Code	Bore	Code	Bore Digit
E01	QTE-700		ACA	Cast Iron SG Iron EN8 AISI 1040 or EN8				-	DE	Valve Mounting-F07, Actuator Mounting-F10		***	Note 3	0	0
E02	QTE-2000		ACB	Cast Iron SG Iron 20MnCr5 AISI 1040 or EN8					EE	Valve Mounting-F10, Actuator Mounting-F10				1	.1
E03	QTE-3000		LAA	Cast Iron SG Iron 20MnCr5 AISI 4140 or EN19					FE	Valve Mounting-F12, Actuator Mounting-F10				2	.2
E04	QTE-3600		LCA	SG Iron SG Iron EN8 AISI 1040 or EN8					GE	Valve Mounting-F14, Actuator Mounting-F10				3	.3
E05	QTE-4000-ER1		ACB	SG Iron SG Iron 20MnCr5 AISI 1040 or EN8					HE	Valve Mounting-F16, Actuator Mounting-F10				4	.4
E06	QTE-6000-ER2		PAM	Cast Iron SG Iron 20MnCr5 AISI 4140 or EN19					JE	Valve Mounting-F25, Actuator Mounting-F10				5	.5
E07	QTE-7500-ER2		PAA	CF8M SG Iron EN8 SS316					KE	Valve Mounting-F30, Actuator Mounting-F10				6	.6
E08	QTE-10000-ER3		PCM	CF8M SG Iron EN8 AISI 1040 or EN8					LE	Valve Mounting-F35, Actuator Mounting-F10				7	.7
E09	QTE-15000-ER3		PMM	CF8M SG Iron 20MnCr5 SS316					YC	Valve Mounting-F07, F10, Actuator Mounting-F10				8	.8
E10	QTE-20000-ER3		QAF	CF8M SG Iron SS316 SS316					YI	Valve Mounting-F05, F07, Actuator Mounting-F10				9	.9
E11	QTE-30000-ER5		QCF	CF8 SG Iron EN8 SS304					14	Valve Mounting-F12(UNC), Actuator Mounting-F10					
E12	QTE-45000-ER5		LBB	CF8 SG Iron 20MnCr5 SS304					JG	Valve Mounting-F25, Actuator Mounting-F14					
			ABB	SG Iron SG Iron EN19 AISI 4140 or EN19					KG	Valve Mounting-F30, Actuator Mounting-F14					
			AAB	Cast Iron SG Iron EN19 AISI 4140 or EN19					LG	Valve Mounting-F35, Actuator Mounting-F14					
			LAB	Cast Iron SG Iron EN8 AISI 4140 or EN19					MG	Valve Mounting-F40, Actuator Mounting-F14					
			LCB	SG Iron SG Iron EN8 AISI 4140 or EN19					**	Note 2					
			WAA	SG Iron SG Iron 20MnCr5 AISI 4140 or EN19					UF	F12(UNC)					
			WCA	WCB SG Iron EN8 AISI 1040 or EN8					UG	F14(UNC)					
			WCB	WCB SG Iron 20MnCr5 AISI 1040 or EN8					UH	F16(UNC)					
			CAA	WCB SG Iron 20MnCr5 AISI 4140 or EN19					UI	F25(UNC)					
			LBA	Cast Iron AB2 EN8 AISI 1040 or EN8					UJ	F30(UNC)					
			SBF	SG Iron SG Iron EN19 AISI 1040 or EN8					UK	F35(UNC)					
			QFF	CF8 AB2 EN19 SS304					UL	F40(UNC)					
			QBF	CF8 SG Iron SS304 SS304					UM	F48(UNC)					
			UFF	CF8 SG Iron EN19 SS304					***	Note 2					
			WBB	CF8 CF8 SS304 SS304											
			1BB	WCB SG Iron EN19 AISI 4140 or EN19											
			***	LCC SG Iron EN19 AISI 4140 or EN19											
			***	Note 1											

Note 1	For more MOC combinations, please contact Q-Tork
Note 2	For multiple Gearbox mounting PCD, please contact Q-Tork
Note 3	Detailed coding is mentioned on Page 62 of this Catalogue
Note 4	For additional accessories, please specify
Note 5	For more specials, please specify

ORIENTATION			HAND WHEEL		ACCESSORIES		IP / LA			SPECIAL		
14	15	16	17	18	19	20	21		22	23	24	25
Y	Y	-	X	X	X	X	1		-	X	X	X

Digit - 14,15		Digit - 16	Digit - 17,18		Digit - 19,20		Digit - 21		Digit-22	Digit - 23,24,25	
Code	Orientation	Code	Code	Handwheel / Chain Wheel	Code	Accessories	Code	IP rating / Padlock	Code	Code	Special
AX	A	-	XX	Without Handwheel	XX	No accessories	1	Enclosure type:- IP65 Without Pad Lock	-	AAH	Hardware SS316, Gear box RAL5015
AB	AB				**	Note 4	2	Enclosure type:- IP67 Without Pad Lock		AAS	Gear Box RAL 9005
AC	AC						3	Enclosure type:- IP68 Without Pad Lock		AAU	Hardware SS316, Gear box RAL9005
AD	AD						4	Enclosure type:- Submersible Without Pad Lock		AAY	Gear box RAL5015
AE	AE						5	Enclosure type:- IP65 With Pad Lock		ABE	Hardware SS304
AF	AF						6	Enclosure type:- IP67 With Pad Lock		ABF	Hardware SS316
AG	AG						7	Enclosure type:- IP68 With Pad Lock		ABJ	Gear box unpainted
AH	AH						8	Enclosure type:- Submersible With Pad Lock		ACJ	Low Temperature Grease
BX	B									XXX	without any specials
BC	BC									***	Note 5
BD	BD										
BE	BE										
BF	BF										
BG	BG										
BH	BH										
CX	C										
CD	CD										
CE	CE										
CF	CF										
CG	CG										
CH	CH										
DX	D										
DE	DE										
DF	DF										
DG	DG										
DH	DH										
EX	E										
EF	EF										
EG	EG										
EH	EH										
FX	F										
FG	FG										
FH	FH										
GX	G										
GH	GH										
HX	H										
A1	ABE										
A2	ABF										
C1	CDG										
C2	CDH										
E1	EFA										
E2	EFB										
G1	GHC										
G2	GHD										
YY	ABEF										
ZZ	CDGH										
XX	NA										

Build Your Code :

BAND FOUR CODE 1																											
QTE	MODEL			MOC-STD.				VM			BORE		DECIMAL	ORIENTATION				HAND WHEEL		ACCESSORIES			IP / LA		SPECIAL		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25			
			-				-								-						-						
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25			
			-				-								-						-						
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25			
			-				-								-						-						
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25			
			-				-								-						-						
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25			
			-				-								-						-						



1710 Series

Quarter-turn Worm Gear Operators with Spline for Electrical & Manual Operations

Q-Tork 1710 series Spline Quarter-turn Worm Gear Operators are designed for Electrical & Manual operation

Application

1710 series Spline Worm Gear Operators are suitable for Ball, Butterfly & Plug Valve.

Features & Benefits

- Splined Bush provides accurate positioning – valve stem & worm wheel
- Option of multiple gear ratios
- Higher M. A. for better performance
- Manual ⇌ Electrical interchangeability
- Submersibility to IP-67 as standard
- Stroke # 0 to 90° ($\pm 5^\circ$) adjustable
- Critical components are heat treated for endurance life
- Reduced frictional losses – worm wheel & Casing
- High performance TRB's and DGBB's
- Buna "N" 'O' ring
- Temperature range -30°C to +120°C
- Environmental friendly high performance grease

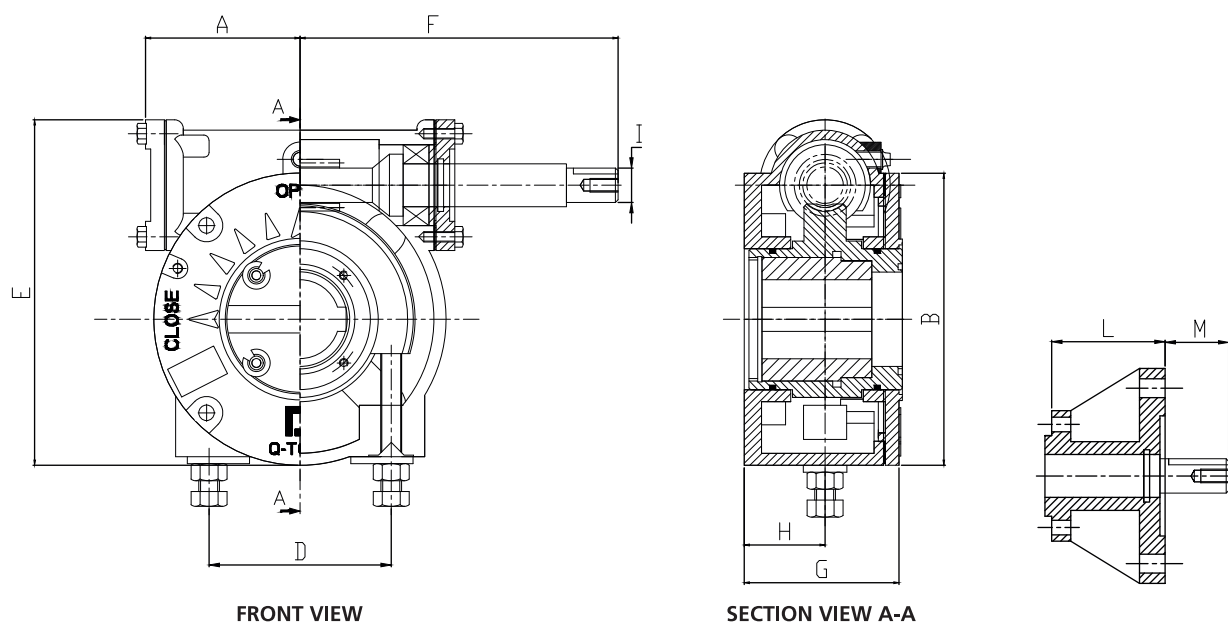
- Light weight with rugged design, Maintenance free
- Standard Gearboxes epoxy primer & powder coated hand wheel

Optional

- MOC # Carbon Steel, Stainless Steel, LCC, LCB
- Chain Wheel operation
- Locking arrangement
- Stainless Steel - Input shaft
- Worm Wheel - Al Bronze

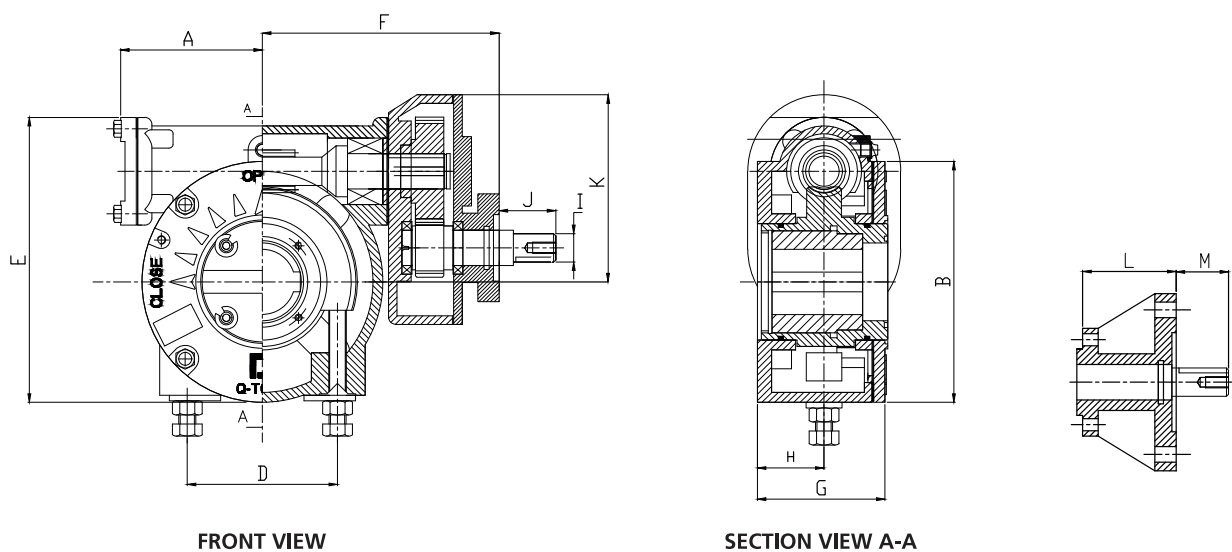
Dimensional Specifications

FIG (1) – Standard Gear Box



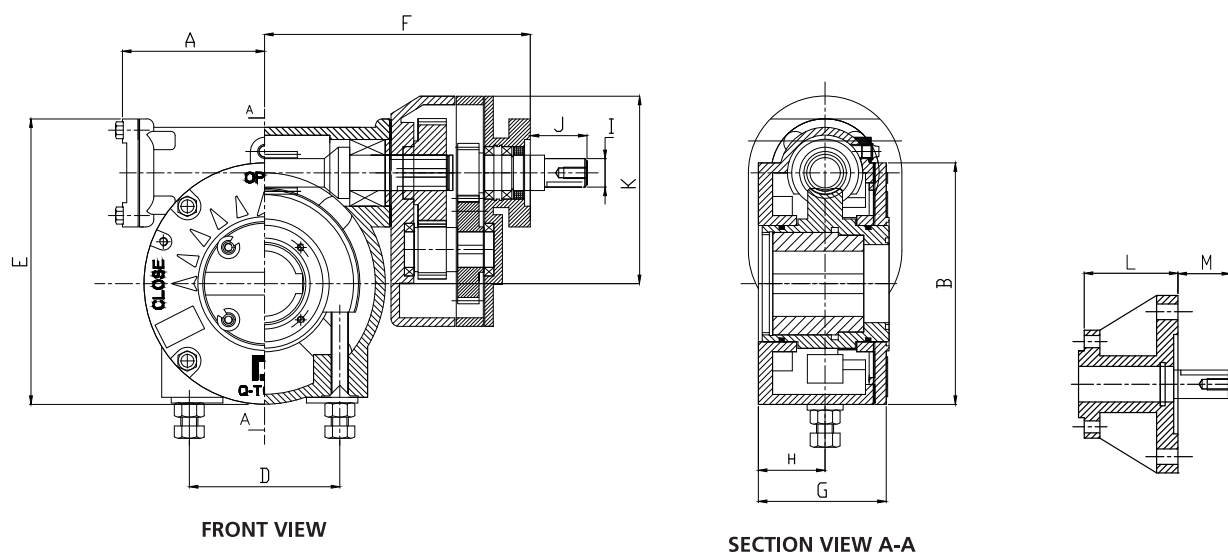
Sr. No.	Model	A	B	D	E	F	G	H	I	L	M
1	QEBC-00	84	104	57	133	136	77	36.5	12	35	50
2	QEBC-01	80	135	77	162.5	149	87	44.5	20	66	50
3	QEBC-02	90	170	100	201	159	90	47	20	66	50
4	QEBC-03	107	205	110	232	188	105	58	20	66	50
5	QEBC-04	118	230	130	258.6	210	113	64	30	66	50
6	QEBC-06	140.8	265	145	314.5	230	141	67	30	85	70
7	QEBC-08	162.8	305	165	350.6	240	148.3	72.5	30	85	70
8	QEBC-10	172.8	324	160	370	250	151.8	76	30	85	70
9	QEBC-13	174	350	200	414	267.5	169	77	40	130	90
10	QEBC-16	196	367	220	440.5	289	178	80	40	130	90
11	QEBC-19	216	410	240	474.5	293	194.3	88	40	130	90
12	QEBC-24	237.8	455.7	295	536.5	329	202.3	88	40	130	90
13	QEBC-32	277.8	525	310	597	350	210.8	95	40	130	90
14	QEBC-42	266	540	330	625.5	381	235	116.5	50	165	90
15	QEBC-54	301	604.7	340	674.5	484	253	130.5	50	165	90
16	QEBC-63	301	619	400	706	429.8	253.3	130.5	50	165	90
17	QEBC-84	321	661	390	740.5	480.8	299	156	50	165	90
18	QEBC-98	345.8	740	460	847	532	280	138.5	60	180	110
19	QEBC-146	390.8	808.7	500	934.5	577	305.8	147	60	180	110

■ FIG (2) – Standard Gear Box with Single Reduction Unit



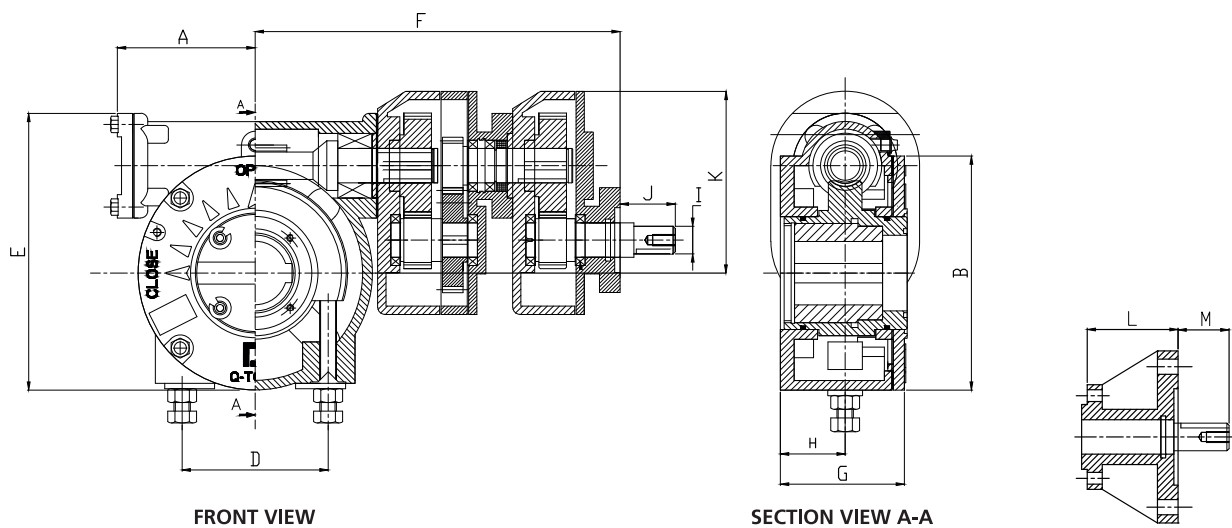
Sr. No.	Model	A	B	D	E	F	G	H	I	J	K	L	M
1	QEBC-02	90	170	100	201	176.5	90	47	20	40	187	66	50
2	QEBC-03	107	205	110	232	193.6	105	58	20	40	200.5	66	50
3	QEBC-04	118	230	130	258.6	204	113	64	30	40	214.5	66	50
4	QEBC-06	140.8	265	145	314.5	265	141	67	30	40	212	85	70
5	QEBC-08	162.8	305	165	350.6	288	148.3	72.5	30	40	228	85	70
6	QEBC-10	172.8	324	160	370	297.4	151.8	76	30	40	238	85	70
7	QEBC-13	174	350	200	414	347.5	169	77	40	50	280	130	90
8	QEBC-16	196	367	220	440.5	369	178	80	40	50	294	130	90
9	QEBC-19	216	410	240	474.5	389	194.3	88	40	50	310	130	90
10	QEBC-24	237.8	455.7	295	536.5	408.1	202.3	88	40	50	338	130	90
11	QEBC-32	277.8	525	310	597	448.1	210.8	95	40	50	368	130	90
12	QEBC-42	266	540	330	625.5	490.3	235	116.5	50	50	414	165	90
13	QEBC-54	301	604.7	340	674.5	525	253	130.5	50	50	439	165	90
14	QEBC-63	301	619	400	706	519.3	253.3	130.5	50	50	455	165	90
15	QEBC-84	321	661	390	740.5	529.5	299	156	50	50	465	165	90
16	QEBC-98	346	740	460	847	625.8	280	138.5	60	50	574	180	110
17	QEBC-146	390.8	808.7	500	934.5	670	305.8	147	60	50	603	180	110

■ FIG (3) – Standard Gear Box with Double Reduction Unit



Sr. No.	Model	A	B	D	E	F	G	H	I	J	K	L	M
1	QEBC-02	90	170	100	201	197.5	90	47	20	40	187	66	50
2	QEBC-03	107	205	110	232	214.6	105	58	20	40	200.5	66	50
3	QEBC-04	118	230	130	258.6	225.4	113	64	30	40	214.5	66	50
4	QEBC-06	140.8	265	145	314.5	293.6	141	67	30	40	212	85	70
5	QEBC-08	162.8	305	165	350.6	316	148.3	72.5	30	40	228	85	70
6	QEBC-10	172.8	324	160	370	325.6	151.8	76	30	40	238	85	70
7	QEBC-13	174	350	200	414	381.9	169	77	40	50	280	130	90
8	QEBC-16	196	367	220	440.5	403.3	178	80	40	50	294	130	90
9	QEBC-19	216	410	240	474.5	423.3	194.3	88	40	50	310	130	90
10	QEBC-24	237.8	455.7	295	536.5	441.2	202.3	88	40	50	338	130	90
11	QEBC-32	277.8	525	310	597	483.3	210.8	95	40	50	368	130	90
12	QEBC-42	266	540	330	625.5	533	235	116.5	50	50	414	165	90
13	QEBC-54	301	604.7	340	674.5	567	253	130.5	50	50	439	165	90
14	QEBC-63	301	619	400	706	567.3	253.3	130.5	50	50	455	165	90
15	QEBC-84	321	661	390	740.5	576.6	299	156	50	50	465	165	90
16	QEBC-98	346	740	460	847	670.8	280	138.5	60	50	574	180	110
17	QEBC-146	390.8	808.7	500	934.5	715	305.8	147	60	50	603	180	110

■ FIG (4) – Standard Gear Box with Triple Reduction Unit



Sr. No.	Model	A	B	D	E	F	G	H	I	J	K	L	M
1	QEBC-02	90	170	100	201	296	90	47	20	40	187	66	50
2	QEBC-03	107	205	110	232	313.4	105	58	20	40	200.5	66	50
3	QEBC-04	118	230	130	258.6	323.9	113	64	30	40	214.5	66	50
4	QEBC-06	140.8	265	145	314.5	431	141	67	30	40	212	85	70
5	QEBC-08	162.8	305	165	350.6	452	148.3	72.5	30	40	228	85	70
6	QEBC-10	172.8	324	160	370	462.2	151.8	76	30	40	238	85	70
7	QEBC-13	174	350	200	414	570	169	77	40	50	280	130	90
8	QEBC-16	196	367	220	440.5	592.2	178	80	40	50	294	130	90
9	QEBC-19	216	410	240	474.5	611.3	194.3	88	40	50	310	130	90
10	QEBC-24	237.8	455.7	295	536.5	631.2	202.3	88	40	50	338	130	90
11	QEBC-32	277.8	525	310	597	672.2	210.8	95	40	50	368	130	90
12	QEBC-42	266	540	330	625.5	778	235	116.5	50	50	414	165	90
13	QEBC-54	301	604.7	340	674.5	806	253	130.5	50	50	439	165	90
14	QEBC-63	301	619	400	706	806.6	253.3	130.5	50	50	455	165	90
15	QEBC-84	321	661	390	740.5	828.4	299	156	50	50	465	165	90
16	QEBC-98	346	740	460	847	981	280	138.5	60	50	574	180	110
17	QEBC-146	390.8	808.7	500	934.5	1025.3	305.8	147	60	50	603	180	110

■ Gear Box Selection Data

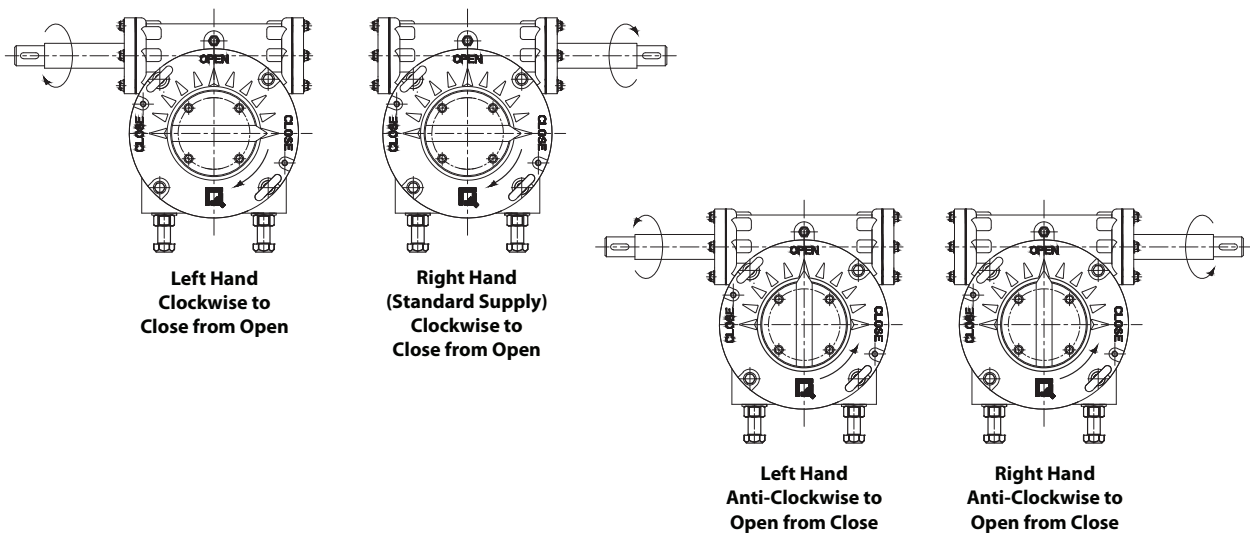
Sr. No.	Model	Max Output Torque		Mechanical Advantage +/-10%	Gear Ratio	Turns to Close	ISO 5211 Valve Mounting	ISO 5211 Actuator Mounting	Maximum Stem Acceptance Dia (mm)	Recommended Hand wheel Dia (mm)
		Nm	Ft-lbf							
1	QEBC-00	480	354	11.8	32:1	8	F07	F10	20	200
	QEBC-00	480	354	12.8	34:1	8.5	F07	F10	20	200
	QEBC-00	480	354	13.1	36:1	9	F07	F10	20	200
2	QEBC-01	960	708	12.2	36:1	9	F07, F10	F10	28	350
	QEBC-01	960	708	13.8	38:1	9.5	F07, F10	F10	28	350
	QEBC-01	960	708	14.5	40:1	10	F07, F10	F10	28	350
3	QEBC-02	1380	1018	16.6	40:1	10	F10, F12	F10	46	350
	QEBC-02	1380	1018	17.4	47:1	11.8	F10, F12	F10	46	350
	QEBC-02/1R1	1380	1018	30.5	80:1	20	F10, F12	F10	46	350
	QEBC-02/2R1	1380	1018	55.8	160:1	40	F10, F12	F10	46	350
	QEBC-02/3R1	1380	1018	100.9	320:1	80	F10, F12	F10	46	350
4	QEBC-03	2400	1770	16.4	46:1	11.5	F10, F12, F14	F10	46	350
	QEBC-03	2400	1770	17.1	50:1	12.5	F10, F12, F14	F10	46	350
	QEBC-03	2400	1770	18.3	55:1	13.8	F10, F12, F14	F10	46	350
	QEBC-03/1R1	2400	1770	30.2	92:1	23	F10, F12, F14	F10	46	350
	QEBC-03/2R1	2400	1770	55.1	184:1	46	F10, F12, F14	F10	46	350
	QEBC-03/3R1	2400	1770	99.7	368:1	92	F10, F12, F14	F10	46	350
5	QEBC-04	3600	2655	18.5	54:1	13.5	F12, F14, F16	F10	65	500
	QEBC-04	3600	2655	15.1	50:1	12.5	F12, F14, F16	F10	65	500
	QEBC-04	3600	2655	16.6	48:1	12	F12, F14, F16	F10	65	500
	QEBC-04/1R1	3600	2655	34.2	108:1	27	F12, F14, F16	F10	65	500
	QEBC-04/2R1	3600	2655	62.5	216:1	54	F12, F14, F16	F10	65	500
	QEBC-04/3R1	3600	2655	113.1	432:1	108	F12, F14, F16	F10	65	500
6	QEBC-06	6000	4425	17.1	50:1	12.5	F14, F16, (F20)	F10, F14	75	600
	QEBC-06	6000	4425	15.4	46:1	11.5	F14, F16, (F20)	F10, F14	75	600
	QEBC-06/1R2	6000	4425	40.1	127.5:1	31.9	F14, F16, (F20)	F10, F14	75	600
	QEBC-06/2R2	6000	4425	93.4	325:1	81.3	F14, F16, (F20)	F10, F14	75	600
	QEBC-06/3R2	6000	4425	215.5	829:1	207.3	F14, F16, (F20)	F10, F14	75	600
7	QEBC-08	7800	5753	18.5	54:1	13.5	F16, F25	F10, F14	80	600
	QEBC-08	7800	5753	16.8	50:1	12.5	F16, F25	F10, F14	80	600
	QEBC-08	7800	5753	16.1	46:1	11.5	F16, F25	F10, F14	80	600
	QEBC-08/1R2	7800	5753	43.4	137.7:1	34.4	F16, F25	F10, F14	80	600
	QEBC-08/2R2	7800	5753	101	351.1:1	87.8	F16, F25	F10, F14	80	600
	QEBC-08/3R2	7800	5753	233.1	895.4:1	223.9	F16, F25	F10, F14	80	600
8	QEBC-10	9720	7169	18.5	56:1	14	F16, F25	F10, F14	85	700
	QEBC-10/1R2	9720	7169	43.4	142.8:1	35.7	F16, F25	F10, F14	85	700
	QEBC-10/2R2	9720	7169	101	364.1:1	91	F16, F25	F10, F14	85	700
	QEBC-10/3R2	9720	7169	233.1	928.6:1	232.1	F16, F25	F10, F14	85	700
9	QEBC-13	12600	9293	18.9	58:1	14.5	F16, F25	F10, F14, F16	95	700
	QEBC-13/1R3	12600	9293	52.2	174:1	43.5	F16, F25	F10, F14, F16	95	700
	QEBC-13/2R3	12600	9293	142.9	522:1	130.5	F16, F25	F10, F14, F16	95	700
	QEBC-13/3R3	12600	9293	387.8	1566:1	391.5	F16, F25	F10, F14, F16	95	700
10	QEBC-16	15600	11506	19.4	60:1	15	F16, F25	F10, F14, F16	95	700
	QEBC-16/1R3	15600	11506	53.5	180:1	45	F16, F25	F10, F14, F16	95	700
	QEBC-16/2R3	15600	11506	146.7	540:1	135	F16, F25	F10, F14, F16	95	700
	QEBC-16/3R3	15600	11506	398.1	1620:1	405	F16, F25	F10, F14, F16	95	700

Sr. No.	Model	Max Output Torque		Mechanical Advantage +/-10%	Gear Ratio	Turns to Close	ISO 5211 Valve Mounting	ISO 5211 Actuator Mounting	Maximum Stem Acceptance Dia (mm)	Recommended Hand wheel Dia (mm)
		Nm	Ft-lbf							
11	QEBC-19	19200	14161	20.2	60:1	15	F25, F30	F10, F14, F16	115	700
	QEBC-19/1R3	19200	14161	55.8	180:1	45	F25, F30	F10, F14, F16	115	700
	QEBC-19/2R3	19200	14161	152.7	540:1	135	F25, F30	F10, F14, F16	115	700
	QEBC-19/3R3	19200	14161	414.5	1620:1	405	F25, F30	F10, F14, F16	115	700
12	QEBC-24	24000	17701	21.4	66:1	16.5	F25, F30	F10, F14, F16	115	700
	QEBC-24/1R3	24000	17701	59.3	198:1	49.5	F25, F30	F10, F14, F16	115	700
	QEBC-24/2R3	24000	17701	162.5	594:1	148.5	F25, F30	F10, F14, F16	115	700
	QEBC-24/3R3	24000	17701	441.2	1782:1	445.5	F25, F30	F10, F14, F16	115	700
13	QEBC-32	32400	23897	22	68:1	17	F25, F30	F10, F14, F16	125	700
	QEBC-32/1R3	32400	23897	60.7	204:1	51	F25, F30	F10, F14, F16	125	700
	QEBC-32/2R3	32400	23897	166.3	612:1	153	F25, F30	F10, F14, F16	125	700
	QEBC-32/3R3	32400	23897	451.4	1836:1	459	F25, F30	F10, F14, F16	125	700
14	QEBC-42	42000	30978	23.2	68:1	17	F25, F30, F35	F16, F25	140	700
	QEBC-42/1R4	42000	30978	85.37	272:1	68	F25, F30, F35	F16, F25	140	700
	QEBC-42/2R4	42000	30978	311.77	1088:1	272	F25, F30, F35	F16, F25	140	700
	QEBC-42/3R4	42000	30978	1128.45	4352:1	1088	F25, F30, F35	F16, F25	140	700
15	QEBC-54	54000	39828	22	68:1	17	F30, F35, F40	F16, F25	180	700
	QEBC-54/1R4	54000	39828	81	272:1	68	F30, F35, F40	F16, F25	180	700
	QEBC-54/2R4	54000	39828	295.7	1088:1	272	F30, F35, F40	F16, F25	180	700
	QEBC-54/3R4	54000	39828	1070.1	4352:1	1088	F30, F35, F40	F16, F25	180	700
16	QEBC-63	62400	46024	23.2	70:1	17.5	F30, F35, F40	F16, F25	180	700
	QEBC-63/1R4	62400	46024	85.4	280:1	70	F30, F35, F40	F16, F25	180	700
	QEBC-63/2R4	62400	46024	311.8	1120:1	280	F30, F35, F40	F16, F25	180	700
	QEBC-63/3R4	62400	46024	1128.4	4480:1	1120	F30, F35, F40	F16, F25	180	700
17	QEBC-84	84000	61955	21.9	64:1	16	F35, F40, F48	F16, F25	210	700
	QEBC-84/1R4	84000	61955	80.6	256:1	64	F35, F40, F48	F16, F25	210	700
	QEBC-84/2R4	84000	61955	294.3	1024:1	256	F35, F40, F48	F16, F25	210	700
	QEBC-84/3R4	84000	61955	1065.2	4096:1	1024	F35, F40, F48	F16, F25	210	700
18	QEBC-98	98400	72576	24	70:1	17.5	F35, F40, F48	F25, F30	210	700
	QEBC-98/1R5	98400	72576	110.4	350:1	87.5	F35, F40, F48	F25, F30	210	700
	QEBC-98/2R5	98400	72576	504	1750:1	437.5	F35, F40, F48	F25, F30	210	700
	QEBC-98/3R5	98400	72576	2280	8750:1	2187.5	F35, F40, F48	F25, F30	210	700
19	QEBC-146	146400	107979	23.4	70:1	17.5	F40, F48	F25, F30	250	700
	QEBC-146/1R5	146400	107979	107.6	350:1	87.5	F40, F48	F25, F30	250	700
	QEBC-146/2R5	146400	107979	491.4	1750:1	437.5	F40, F48	F25, F30	250	700
	QEBC-146/3R5	146400	107979	2223	8750:1	2187.5	F40, F48	F25, F30	250	700

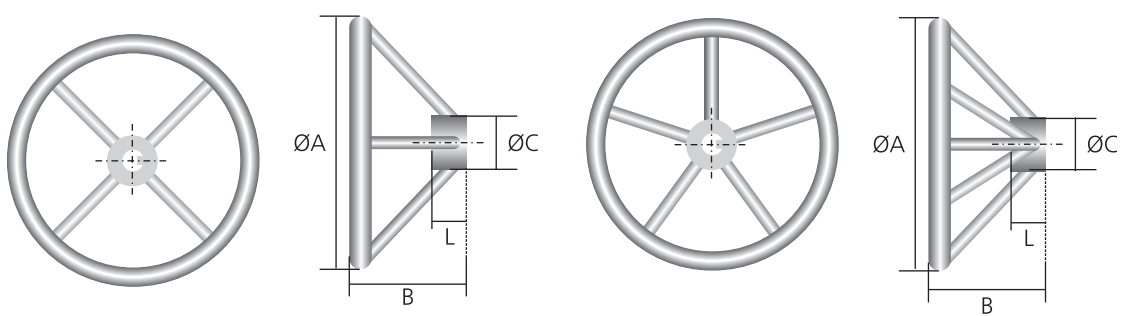
Material Specifications

Sr. No	Job Description	Material Description	Standards
1	Housing	SG Iron-500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
2	Cover	SG Iron-500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
3	Worm Wheel	SG Iron-600/3EN19	IS 1865 / ASTM A536 80-60-03 / GGG-60
4	Worm Shaft	EN19	45C8 / SAE 1040 / CK45
5	Spline Bush	SG Iron 600/3	IS 1865 / ASTM A536 80-60-03 / GGG-60
6	Reduction Housing	SG Iron-500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
7	Reduction Cover	SG Iron 500/7	IS 1865 / ASTM A536 80-55-06 / GGG-50
8	Spur Pinion	EN19	C55 / SAE 4140 / 42CrMo4
9	Spur Gear	EN19	C55 / SAE 4140 / 42CrMo4
10	Bearing	Standard	—

Input Shaft and Drive Orientation



Handwheel with Keyway



* Optional: Hand wheel with Dowel Hole

Sr.No	Handwheel Dia-A	Height B	Bush OD Dia-C	Bush length-L	No.of Spokes
1	200	80	25	25	4
		50	35	35	3
2	350	110	40	30	4
		100	30	25	
3	500	110	65	30	4
4	600	130	65	30	5
5	700	150	65	30	5

Ordering Information

QEBC	Model			MOC-STD					VM & AM		BORE		DECIMAL				
1	2	3	4	5		6		7	8	9	10	11	12	13			
C	0	5	-	M		B		B	-	H	X	A	L	0			
Digit - 1,2,3			Digit -4	Digit - 5,6,7					Digit -8	Digit - 9,10		Digit - 11,12		Digit - 13			
Code	Model	-	MBB	ASTM A536 (80-55-06) ASTM A536 (80-60-03) ASTM A536 (80-60-03) AISI 4140 or EN19					Code	Code	Mounting		Code	Bore	Code	Bore	Digit
C01	QEBC-00		MBF	ASTM A536 (80-55-06) ASTM A536 (80-60-03) ASTM A536 (80-60-03) SS304					-	DX	Valve Mounting-F07, Actuator Mounting-NA			Note 3	0	0	
C02	QEBC-01		MBM	ASTM A536 (80-55-06) ASTM A536 (80-60-03) ASTM A536 (80-60-03) SS316						EX	Valve Mounting-F10, Actuator Mounting-NA					1	.1
C03	QEBC-02		MBH	ASTM A536 (80-55-06) ASTM A536 (80-60-03) ASTM A536 (80-60-03) SS431						FX	Valve Mounting-F12, Actuator Mounting-NA					2	.2
C04	QEBC-03		TPN	EN 1982:2017 Gr. CC492K BS2874 PB 102 BS2874 PB 102 NES 833 PART2 CLASS2						GX	Valve Mounting-F14, Actuator Mounting-NA					3	.3
C05	QEBC-04		RII	BS 2789 SGI 500-7 FORGED STEEL BS 970 316 FORGED STEEL BS 970 316 FORGED STEEL BS 970 316						HX	Valve Mounting-F16, Actuator Mounting-NA					4	.4
C06	QEBC-06		***	Note 1						JX	Valve Mounting-F25, Actuator Mounting-NA					5	.5
C07	QEBC-08		PAA	CF8M SG Iron EN8 SS316						KX	Valve Mounting-F30, Actuator Mounting-NA					6	.6
C08	QEBC-10		PCM	CF8M SG Iron EN8 AISI 1040 or EN8						LX	Valve Mounting-F35, Actuator Mounting-NA					7	.7
C09	QEBC-13		PMM	CF8M SG Iron 20MnCr5 SS316						MX	Valve Mounting-F40, Actuator Mounting-NA					8	.8
C10	QEBC-16		QAF	CF8M SG Iron SS316 SS316						NX	Valve Mounting-F48, Actuator Mounting-NA					9	.9
C11	QEBC-19		QCF	CF8 SG Iron EN8 SS304						DE	Valve Mounting-F07, Actuator Mounting-F10						
C12	QEBC-24		LBB	CF8 SG Iron 20MnCr5 SS304						EE	Valve Mounting-F10, Actuator Mounting-F10						
C13	QEBC-32		ABB	SG Iron SG Iron EN19 AISI 4140 or EN19						FE	Valve Mounting-F12, Actuator Mounting-F10						
C14	QEBC-42		AAB	Cast Iron SG Iron EN19 AISI 4140 or EN19						FG	Valve Mounting-F12, Actuator Mounting-F14						
C15	QEBC-54		LAB	Cast Iron SG Iron EN8 AISI 4140 or EN19						FH	Valve Mounting-F12, Actuator Mounting-F16						
C16	QEBC-63		LCB	SG Iron SG Iron EN8 AISI 4140 or EN19						GE	Valve Mounting-F14, Actuator Mounting-F10						
C17	QEBC-84		WAA	SG Iron SG Iron 20MnCr5 AISI 4140 or EN19						GG	Valve Mounting-F14, Actuator Mounting-F14						
C18	QEBC-98		WCA	WCB SG Iron EN8 AISI 1040 or EN8						HE	Valve Mounting-F16, Actuator Mounting-F10						
C19	QEBC-146		WCB	WCB SG Iron 20MnCr5 AISI 1040 or EN8						HG	Valve Mounting-F16, Actuator Mounting-F14						
C23	QEBC-02-1R1		CAA	WCB SG Iron 20MnCr5 AISI 4140 or EN19						HH	Valve Mounting-F16, Actuator Mounting-F16						
C24	QEBC-03-1R1		LBA	Cast Iron AB2 EN8 AISI 1040 or EN8						JE	Valve Mounting-F25, Actuator Mounting-F10						
C25	QEBC-04-1R1		SBF	SG Iron SG Iron EN19 AISI 1040 or EN8						JG	Valve Mounting-F25, Actuator Mounting-F14						
C26	QEBC-06-1R2		QFF	CF8 AB2 EN19 SS304						JH	Valve Mounting-F25, Actuator Mounting-F16						
C27	QEBC-08-1R2		QBF	CF8 SG Iron SS304 SS304						JJ	Valve Mounting-F25, Actuator Mounting-F25						
C28	QEBC-10-1R2		UFF	CF8 SG Iron EN19 SS304						KE	Valve Mounting-F30, Actuator Mounting-F10						
C29	QEBC-13-1R3		WBB	CF8 CF8 SS304 SS304						KG	Valve Mounting-F30, Actuator Mounting-F14						
C30	QEBC-16-1R3		1BB	WCB SG Iron EN19 AISI 4140 or EN19						KH	Valve Mounting-F30, Actuator Mounting-F16						
C31	QEBC-19-1R3		***	LCC SG Iron EN19 AISI 4140 or EN19						KJ	Valve Mounting-F30, Actuator Mounting-F25						
C32	QEBC-24-1R3		***	Note 1						LH	Valve Mounting-F35, Actuator Mounting-F16						
C33	QEBC-32-1R3									LJ	Valve Mounting-F35, Actuator Mounting-F25						
C34	QEBC-42-1R4									LK	Valve Mounting-F35, Actuator Mounting-F30						
C35	QEBC-54-1R4									MH	Valve Mounting-F40, Actuator Mounting-F16						
C36	QEBC-63-1R4									MJ	Valve Mounting-F40, Actuator Mounting-F25						
C37	QEBC-84-1R4									MK	Valve Mounting-F40, Actuator Mounting-F30						
C38	QEBC-98-1R5									NH	Valve Mounting-F48, Actuator Mounting-F16						
C39	QEBC-146-1R5									NJ	Valve Mounting-F48, Actuator Mounting-F25						
C43	QEBC-02-2R1									NK	Valve Mounting-F48, Actuator Mounting-F30						
C44	QEBC-03-2R1									**	Note 2						
C45	QEBC-04-2R1																
C46	QEBC-06-2R2																
C47	QEBC-08-2R2																
C48	QEBC-10-2R2																
C49	QEBC-13-2R3																
C50	QEBC-16-2R3																
C51	QEBC-19-2R3																
C52	QEBC-24-2R3																
C53	QEBC-32-2R3																
C54	QEBC-42-2R4																
C55	QEBC-54-2R4																
C56	QEBC-63-2R4																
C57	QEBC-84-2R4																
C58	QEBC-98-2R5																
C59	QEBC-146-2R5																
C63	QEBC-02-3R1																
C64	QEBC-03-3R1																
C65	QEBC-04-3R1																
C66	QEBC-06-3R2																
C67	QEBC-08-3R2																
C68	QEBC-10-3R2																
C69	QEBC-13-3R3																
C70	QEBC-16-3R3																
C71	QEBC-19-3R3																
C72	QEBC-24-3R3																
C73	QEBC-32-3R3																
C74	QEBC-42-3R4																
C75	QEBC-54-3R4																
C76	QEBC-63-3R4																
C77	QEBC-84-3R4																
C78	QEBC-98-3R5																
C79	QEBC-146-3R5																

Note 1

For more MOC combinations, please contact Q-Tork

Note 2

For multiple Gearbox mounting PCD, please contact Q-Tork

Note 3

Detailed coding is mentioned on Page 62 of this Catalogue

Note 4

For additional accessories, please specify

Note 5

For more specials, please specify

ORIENTATION			HAND WHEEL		ACCESSORIES		IP / LA			SPECIAL				
14	15	16	17		18		19	20	21		22	23	24	25
X	X	-	K		H		X	X	6		-	X	X	X
Digit - 14,15		Digit - 16	Digit - 17,18			Digit - 19,20		Digit - 21			Digit-22	Digit - 23,24,25		
Code	Orientation	Code	Code	Handwheel / Chain Wheel		Code	Accessories	Code	IP rating / Padlock			Code	Code	Special
AX	A	-	KB	MS Fabricated H/W:- Dia 200mm Key Type		XX	No accessories	2	Enclosure type:- IP67 Without Pad Lock			—	AAH	Hardware SS316, Gear box RAL5015
AB	AB		KE	MS Fabricated H/W:- Dia 350mm Key Type				Note 4	3	Enclosure type:- IP68 Without Pad Lock			AAS	Gear Box RAL 9005
AC	AC		KH	MS Fabricated H/W:- Dia 500mm Key Type				4	Enclosure type:- Submersible Without Pad Lock			AAU	Hardware SS316, Gear box RAL9005	
AD	AD		KK	MS Fabricated H/W:- Dia 600mm Key Type				6	Enclosure type:- IP67 With Pad Lock			AAY	Gear box RAL5015	
AE	AE		KN	MS Fabricated H/W:- Dia 700mm Key Type				7	Enclosure type:- IP68 With Pad Lock			ABE	Hardware SS304	
AF	AF		XX	Without Handwheel				8	Enclosure type:- Submersible With Pad Lock			ABF	Hardware SS316	
AG	AG							7	Enclosure type:- IP68 With Pad Lock			ABJ	Gear box unpainted	
AH	AH							8	Enclosure type:- Submersible With Pad Lock			ACJ	Low Temperature Grease	
BX	B											XXX	without any specials	
BC	BC											***	Note 5	
BD	BD													
BE	BE													
BF	BF													
BG	BG													
BH	BH													
CX	C													
CD	CD													
CE	CE													
CF	CF													
CG	CG													
CH	CH													
DX	D													
DE	DE													
DF	DF													
DG	DG													
DH	DH													
EX	E													
EF	EF													
EG	EG													
EH	EH													
FX	F													
FG	FG													
FH	FH													
GX	G													
GH	GH													
HX	H													
A1	ABE													
A2	ABF													
C1	CDG													
C2	CDH													
E1	EFA													
E2	EFB													
G1	GHC													
G2	GHD													
YY	ABEF													
ZZ	CDGH													
XX	NA													

Build Your Code :

QEBC	MODEL			MOC-STD.				VM		BORE		DECIMAL	ORIENTATION		HAND WHEEL	ACCESSORIES		IP / LA	SPECIAL					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				

Actual Product Photos



*Bronze MOC



234 Series | Manual Quarter Turn Worm Gear Operators with Rim Effort

Q-Tork 234 series has been designed to meet stringent of the industry to provide the levels of durability and reliability. This gear box designed to operate with less manual effort.

Application

234 Series quarter turn worm gearboxes are used to operate ball valve, Butterfly valve, plug valve and dampers.

Features & Benefits

- Graded CI Housing & Cover
- High performance axial bearing
- Powder coated Handwheels for easy & smooth operation
- Temperature Range -30°C to +120°C
- Environmental friendly high performance grease
- Dowel pin arrangement at mounting side
- Manual effort less than 360N
- Stroke # 90° (±5°)
- Left hand & Right hand input shaft projection
- Alloy steel-worm & shaft

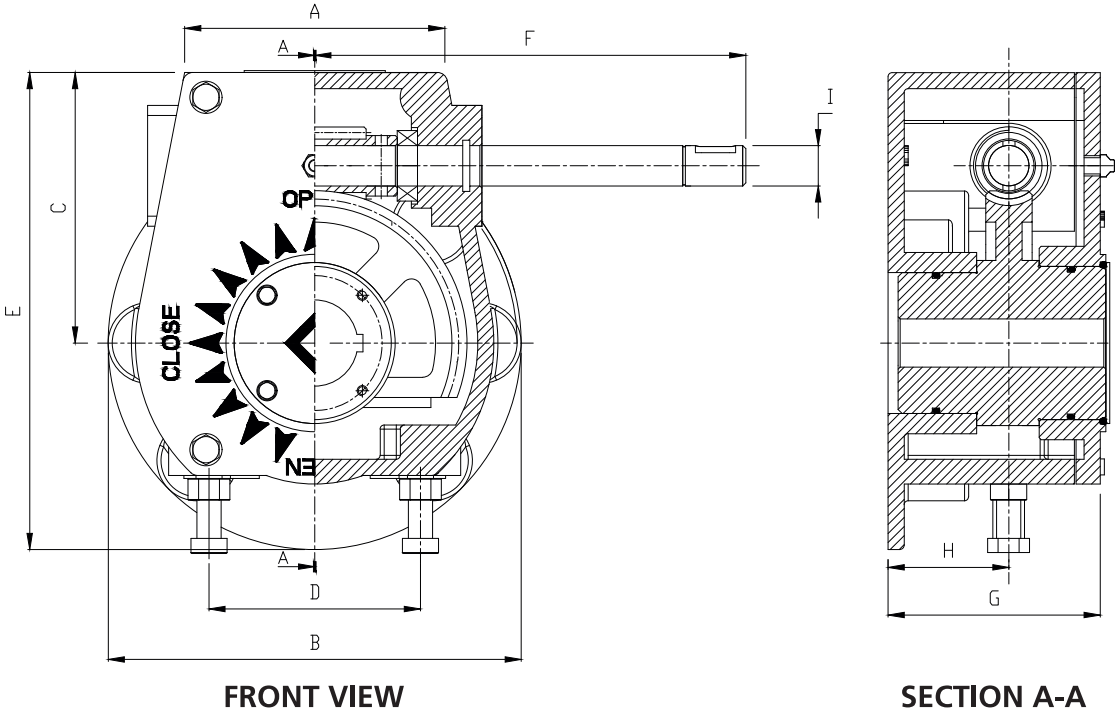
Optional

- Offering MOC:
 - ▶ Housing/ Cover- Carbon steel, Ductile Iron, Stainless Steel
 - ▶ Quadrant - Al. Bronze
- Stainless steel- Input shaft, handwheel and hardware
- Lubricants in temperature variants can be provided.
- Submersibility to IP67 & IP68
- Limit switch arrangement
- Splined bush provide accurate positioning- valve stem & worm wheel

An ISO 9001:2015 Company

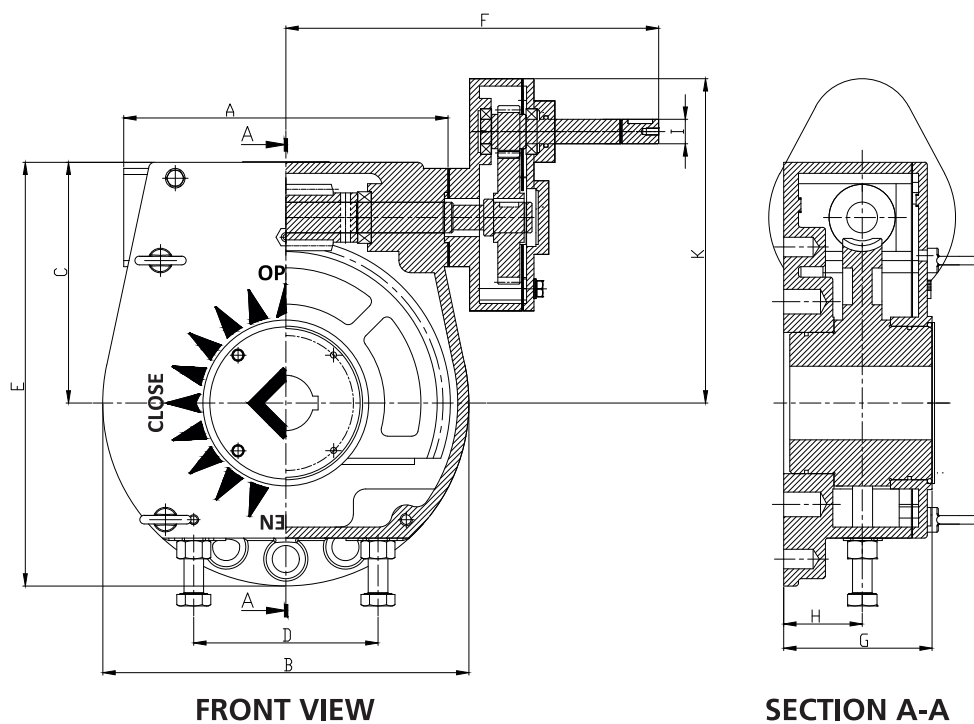
■ Dimensional Specifications

■ FIG (1)



Sr. No.	MODELS	A	B	C	D	E	F	G	H	I	No. Of Dowel Hole
1	QERG-03	83	98 146	71.5	52	107	160	55	32	12	2
2	QREG-06	106	112 146	84	65	127	153	64	35	12	2
3	QERG-09	115	149	98	70	141	162	71	41	20	2
4	QERG-12	150	194	27	91	183	210	92	53	20	2
5	QERG-18	14	205	134	105	200	214	105	60	20	2
6	QERG-24	169	205	136	115	216	251	106	60	20	2
7	QERG-36	196	300	167	120	317	255.5	116	61	20	2

■ FIG (2)



Sr. No.	MODELS	A	B	C	D	E	F	G	H	I	J	No. Of Dowel Hole
8	QERG-36/RR2	209	242	167	120	252	287	115	60	20	235	4
9	QERG-54/RR2	206	282	181	140	282	300	119	65.5	20	253	4
10	QERG-66/RR2	240	298	196	150	306	305	117	64	20	264	4
11	QERG-84/RR3	248	38	203	155	315	333	139	81	20	295	4
12	QERG-102/RR3	260	344	235	170	361	339	137	79	20	322	4
13	QERG-120/RR3	306	367	247	185	356.5	365	142	80	20	332	4

Gear Box Selection Data

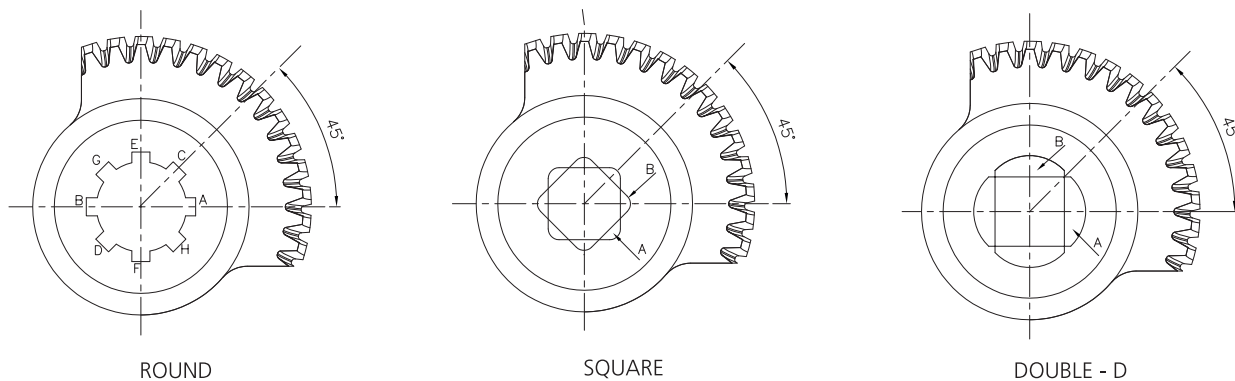
Sr. No	Model	Output Torque		Gear Ratio	MA ±10 %	ISO 5211 Valve Mounting	Max Stem Acceptance Dia (mm)	Max Stem Acceptance Dia with Spline (mm)	Approx. unit weight (Kgs)	Recommended Handwheel Dia(mm)
		Nm	Ft-lbf							
1	QREG-03	250	184	35:1	10.1	F05, F07	22	16	2.9	200
						F10, F12	35	22	3.9	
2	QREG-06	500	369	42:1	11.3	F05, F07	22	16	4.6	300
						F10, F12	35	22	5.2	
3	QREG-09	750	553	46:1	11.9	F07, F10, F12	38	23	7.5	350
4	QREG-12	1000	783	50:1	13.1	F10, F12, F14	45	32	11.0	450
5	QREG-18	1500	1106	52:1	15.5	F10, F12, F14, F16	50	35	17.7	550
6	QREG-24	2000	1475	60:1	18.0	F10, F12, F14, F16	60	40	18.5	650
7	QREG-36	2000	1475	68:1	20.5	F12, F14, F16, F25	70	50	28.5	650
8	QREG-36/RR2	3000	2213	204:1	55.1	F12, F14, F16	70	50	33.5	350
9	QREG-54/RR2	4500	3319	216:1	56.5	F16, F25	85	65	49.3	450
10	QREG-66/RR2	5500	4057	192:1	57.2	F16, F25	85	65	52	550
11	QREG-84/RR3	7000	5163	255:1	68.7	F16, F25, F30	95	70	66.2	600
12	QREG-102/RR3	8500	6269	330:1	85.2	F16, F25, F30	95	70	81.2	600
13	QREG-120/RR3	10000	7376	330:1	85.9	F25, F30	95	70	87.6	650

* Spline bush is not standard. Please specify if required

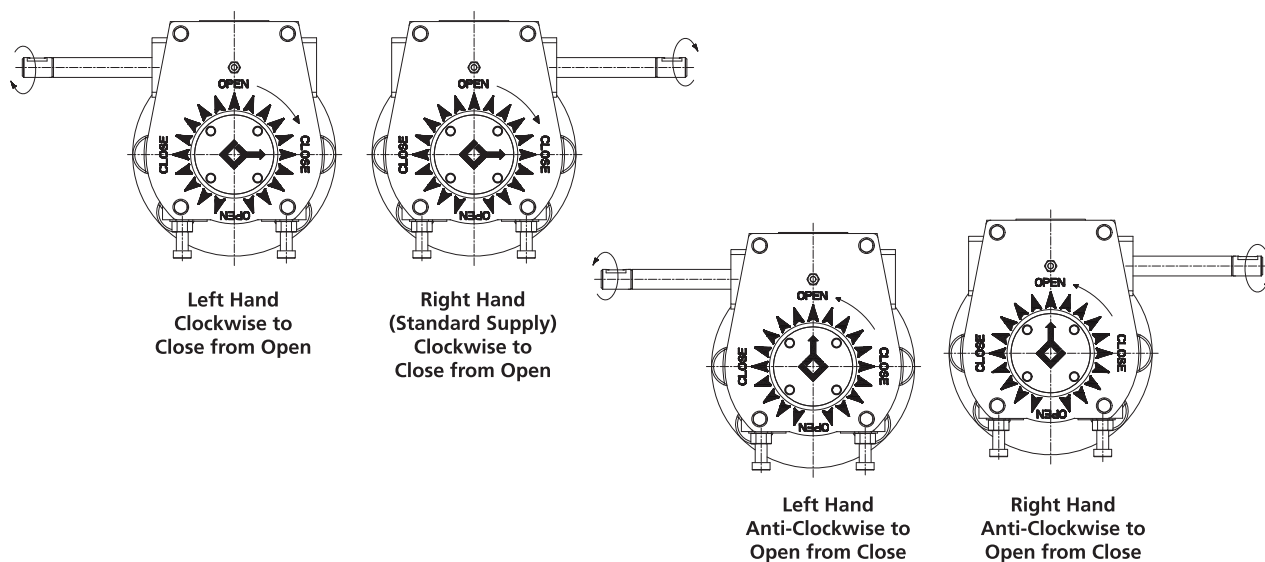
Material Specifications

Sr. No	Job Description	Material Description	Standard
1	Housing	Ductile Iron	ASTM A536 65-45-12/SG-450/10
2	Cover	Ductile Iron	ASTM A536 65-45-12/SG-450/10
3	Worm Wheel	Ductile Iron	ASTM A536 80-55-06/SG-500/7
4	Worm	EN8/EN19	45C8/ AISI-1040/AISI 4140/CK45/C55/ASTM A-29/A-29M
5	Input Shaft	EN8/EN19	45C8/ AISI-1040/AISI 4140/CK45/C55/ASTM A-29/A-29M
6	Drive Shaft	EN8/EN19	45C8/AISI-1040/AISI 4140/CK45/C55/ASTM A-29/A-29M
7	Spur Pinion	EN19	C55/SAE 4140/ 42CrM04/ASTM A-29/A-29M
8	Spur Gear	EN19	C55/SAE 4140/ 42CrM04/ASTM A-29/A-29M
		Ductile Iron	ASTM A536 80-55-06/SG-450/10

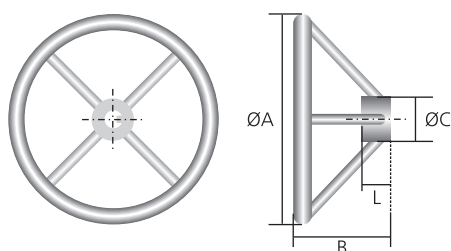
Worm Wheel Orientation Options as seen from Top View at Closed Position



Input Shaft and Drive Orientation



Handwheel with Keyway



* Optional: Hand wheel with Dowel Hole

Sr. No	Handwheel Dia-A	Height B	Bush OD Dia-C	Bush Length-L	No. Of Spokes
1	200	80	25	25	4
2	300	100	30	25	4
3	350	110	40	30	4
4	450	145	50	30	4
5	550	120	65	30	4
6	650	140	65	30	5

Ordering Information

QREG	Model				MOC-STD				VM		BORE		DECIMAL
1	2	3	4		5	6	7	8	9	10	11	12	13
E	0	3	-		A	C	A	-	H	E	C	Q	0

Digit - 1,2,3		Digit -4		Digit - 5,6,7				Digit -8		Digit - 9,10		Digit - 11,12		Digit - 13	
Code	Model	Code	Code	MOC - Housing, Cover	Worm Wheel	Worm	Input Shaft	Code	Code	Mounting		Code	Bore	Code	Bore Digit
G01	QREG-03	—	AAA	ASTM A536 (65-45-12) ASTM A536 (80-60-03) EN19 EN19				-	CX	F05		***	Note 3	0	0
G02	QREG-06		***	Note 1					DX	F07				1	.1
G03	QREG-09								EX	F10				2	.2
G04	QREG-12								FX	F12				3	.3
G05	QREG-18								GX	F14				4	.4
G06	QREG-24								HX	F16				5	.5
G07	QREG-36								JX	F25				6	.6
G08	QREG-36-RR2								KX	F30				7	.7
G09	QREG-54-RR2								UC	F05(UNC)				8	.8
G10	QREG-66-RR2								UD	F07(UNC)				9	.9
G11	QREG-84-RR3								UE	F10(UNC)					
G12	QREG-102-RR3								UF	F12(UNC)					
G13	QREG-120-RR3								UG	F14(UNC)					
									UH	F16(UNC)					
									UI	F25(UNC)					
									UJ	F30(UNC)					
									***	Note 2					

Note 1	For more MOC combinations, please contact Q-Tork
Note 2	For multiple Gearbox mounting PCD, please contact Q-Tork
Note 3	Detailed coding is mentioned on Page 62 of this Catalogue
Note 4	For additional type and size, Please contact Q-Tork
Note 5	For additional accessories, please specify
Note 6	For more specials, please specify

ORIENTATION			HAND WHEEL			ACCESSORIES		IP / LA		SPECIAL		
14	15	16	17	18	19	20	21	22	23	24	25	
Y	Y	-	X	X	X	X	1	-	X	X	X	

Digit - 14,15		Digit - 16		Digit - 17,18		Digit - 19,20		Digit - 21		Digit-22		Digit - 23,24,25	
Code	Orientation	Code	Code	Handwheel / Chain Wheel	Code	Accessories	Code	IP rating / Padlock	Code	Code	Special		
AX	A	-	KB	MS Fabricated H/W:- Dia 200mm Key Type	BF	Bottom side Bracket & Adaptor	1	Enclosure type:- IP65 Without Pad Lock	-	AAA	NCP shaft & Hardware SS304		
AB	AB		KD	MS Fabricated H/W:- Dia 300mm Key Type	BG	bottom side Adaptor	2	Enclosure type:- IP67 Without Pad Lock		AAH	Hardware SS316, Gear box RAL5015		
AC	AC		KE	MS Fabricated H/W:- Dia 350mm Key Type	SA	Shaft Extension	3	Enclosure type:- IP68 Without Pad Lock		AAL	Gear box and handwheel RAL3001		
AD	AD		KG	MS Fabricated H/W:- Dia 450mm Key Type	TA	With Limit Switch, Bracket, Adapter	4	Enclosure type:- Submersible Without Pad Lock		AAQ	Dowel hole arrangement		
AE	AE		KJ	MS Fabricated H/W:- Dia 550mm Key Type	TB	Limit Switch Arrangement	5	Enclosure type:- IP65 With Pad Lock		AAS	Gear Box RAL 9005		
AF	AF		KK	MS Fabricated H/W:- Dia 600mm Key Type	TD	Position Indicator Scale	6	Enclosure type:- IP67 With Pad Lock		AAU	Hardware SS316, Gear box RAL9005		
AG	AG		KM	MS Fabricated H/W:- Dia 650mm Key Type	XX	No accessories	7	Enclosure type:- IP68 With Pad Lock		AAY	Gear box RAL5015		
AH	AH		XX	Without Handwheel	**	Note 5	8	Enclosure type:- Submersible With Pad Lock		ABE	Hardware SS304		
BX	B		**	Note 4						ABF	Hardware SS316		
BC	BC									ABJ	Gear box unpainted		
BD	BD									ABQ	Both Side Shaft		
BE	BE									ABS	Solid Hand wheel		
BF	BF									ACJ	Low Temperature Grease		
BG	BG									XXX	without any specials		
BH	BH									***	Note 6		
CX	C												
CD	CD												
CE	CE												
CF	CF												
CG	CG												
CH	CH												
DX	D												
DE	DE												
DF	DF												
DG	DG												
DH	DH												
EX	E												
EF	EF												
EG	EG												
EH	EH												
FX	F												
FG	FG												
FH	FH												
GX	G												
GH	GH												
HX	H												
A1	ABE												
A2	ABF												
C1	CDG												
C2	CDH												
E1	EFA												
E2	EFB												
G1	GHC												
G2	GHD												
YY	ABEF												
ZZ	CDGH												
XX	NA												

Build Your Code :

QREG	MODEL			MOC-STD.				VM		BORE		DECIMAL		ORIENTATION		HAND WHEEL		ACCESSORIES IP / LA		SPECIAL				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			-				-								-					-				



519 Series | Multi-turn Bevel Gear Operators

Q-Tork 519 series bevel gear operators are suitable for manual as well as motorized operation

Application

519 Series operator are multi-turn bevel gear boxes are used in operation of Gate Valve, Globe Valve, Sluice Valve and penstock Valve.

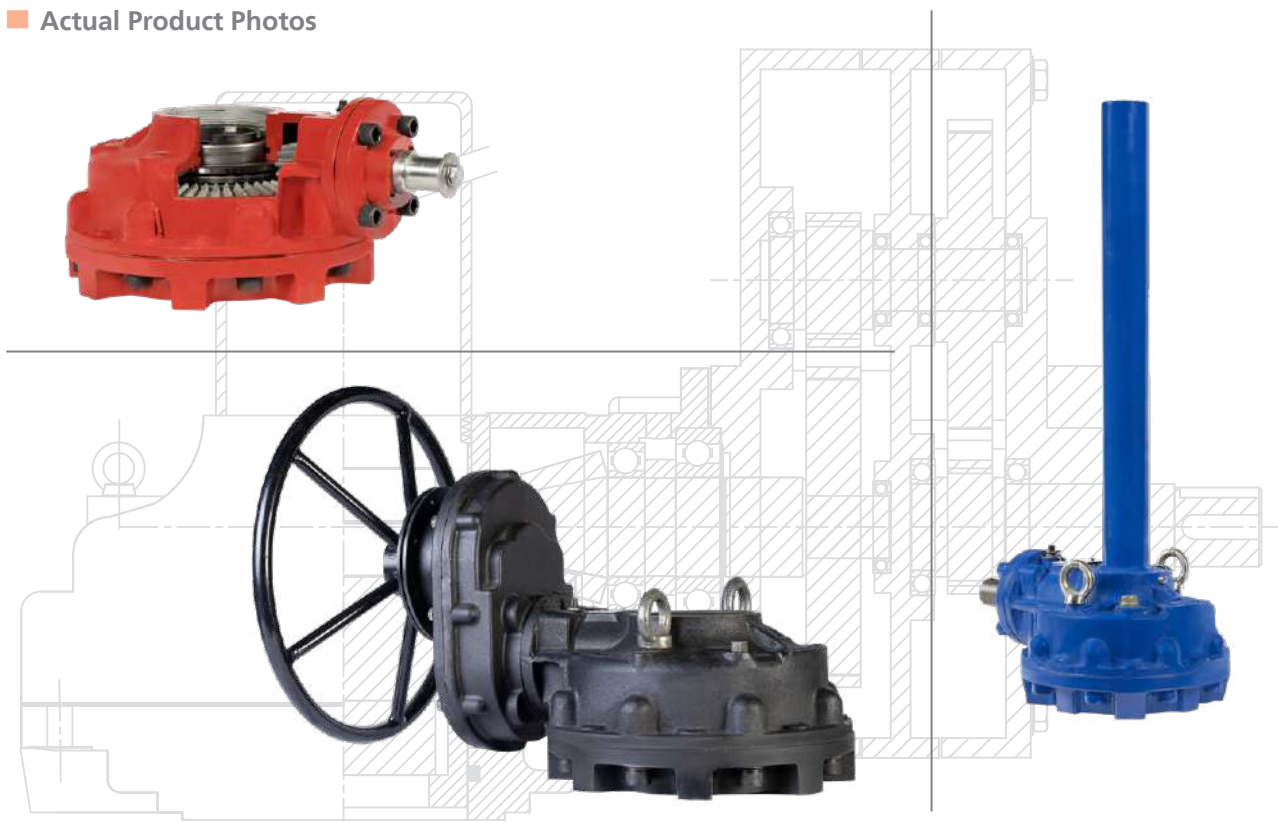
Features & Benefits

- Ductile Iron - Housing, Base and End cap
- Alloy steel - Gear and Pinion
- All critical components are heat treated to ensure longer serviceable life
- High Performance bearing
- Environmental friendly high performance grease
- Powder coated Handwheels for smooth and easy operation
- Epoxy Primer Coating
- Removable Stem Nut to facilitate spindle machining thereby enabling customer stocking
- Stem Nut - Al. Bronze
- Temperature range -30°C to +120°C
- Light weight with rugged design

Optional

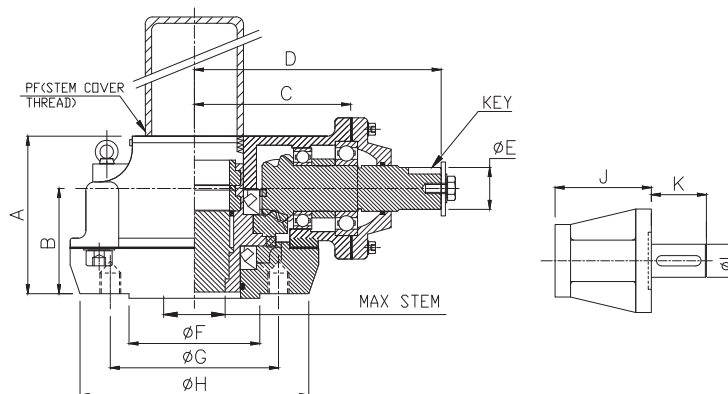
- Housing Materials - Carbon Steel, Cast Iron
- Locking arrangement
- Submersibility to IP67
- Grease High up to +260°C
- Grease Low up to -45°C

Actual Product Photos



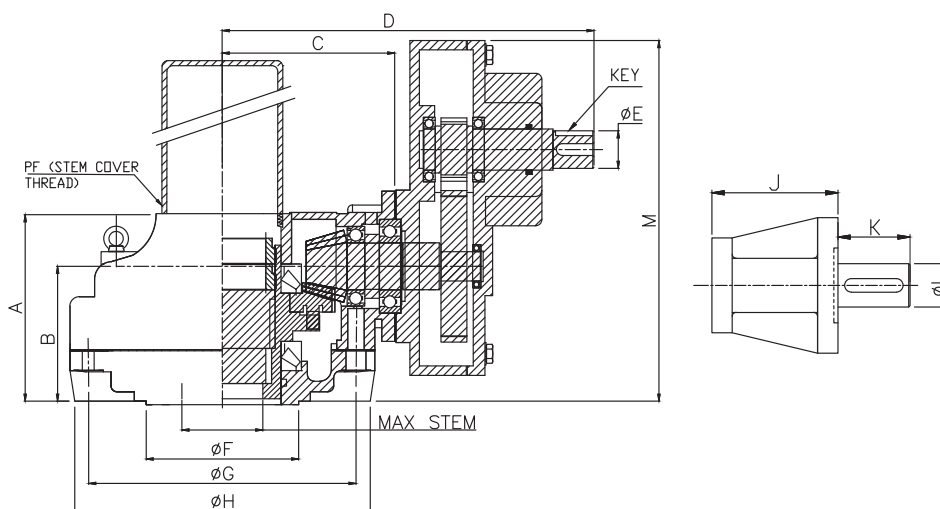
Dimensional Specifications

FIG-(1) Standard Gear Box



Sr. No	Models	A	B	C	D	ØE	Key	Stem Cover Thread	ØF	ØG	ØH	J	K	ØL	Actuator Mounting Flange
1	QTB-220	95	71	90	164	26	8x7	PF 2"	70	102	125	70	50	20	F10
2	QTB-370	106	76.6	104.5	178.5	26		PF 2 1/2"	70 85	102 125	125 150	70	50	20	F10
3	QTB-600	121	81	119.5	198.5	32		PF 2 1/2"	85 100	125 140	150 175	85	70	30	F14
4	QTB-980	140	98	130	210	32	10x8	PF 3"	100 130	140 165	175 210	85	70	30	F14
5	QTB-1500	160	115	147	231	32		PF 3 1/2"	130 140	165 205	210 250	85	70	30	F14
6	QTB-2500	167	119	167	252	32		PF 4"	200	254	300	85	70	30	F14
7	QTB-3500	196	138	217.5	319	38	12x8	PF 5"	230	298	350	106.5	90	40	F16
8	QTB-5200	240	165.5	248	351	38		PF 6"	230	298	350	120	90	40	F16
9	QTB-7800	253	182.5	286	387	38		PF 6"	260	356	415	120	90	40	F16
10	QTB-13000	279	201	321	441	50	16x10	PF 8"	300	406	475	120	90	40	F16

■ FIG(2)-Standard Gear Box with Single Reduction Unit



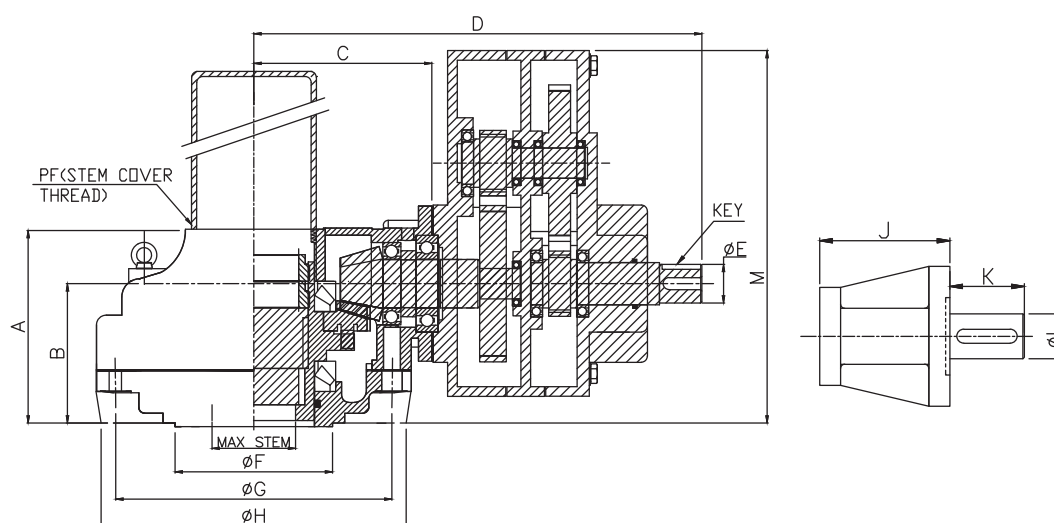
■ Spur Reduction Unit

Sr. No	Models	A	B	C	D	ØE	Key	Stem Cover Thread	ØF	ØG	ØH	M	J	K	ØL	Actuator Mounting Flange
1	QTB-370/1S0	106	76.6	104.5	254	20	8x7	PF 2 1/2"	70 85	102 125	125 150	170	70	50	20	F10
2	QTB-600/1S1	121	81	119.5	291	32		PF 2 1/2"	85 100	125 140	150 175	256.5	85	70	30	F14
3	QTB-980/1S1	140	98	130	301.5	32		PF 3"	100 130	140 165	175 210	275	85	70	30	F14
4	QTB-1500/1S2	160	115	147	317	32		PF 3 1/2"	130 140	165 205	210 250	308	85	70	30	F14
5	QTB-2500/1S2	167	119	167	337	32		PF 4"	200	254	300	311	85	70	30	F14
6	QTB-3500/1S3	196	138	217.5	437	38		PF 5"	230	298	350	370	106.5	90	40	F16
7	QTB-5200/1S3	240	165.5	248	467	38		PF 6"	230	298	350	390	120	90	40	F16
8	QTB-7800/1S4	253	182.5	286	514	38		PF 6"	260	356	415	453	120	90	40	F16
9	QTB-13000-1S5	279	201	321	577	50	16x10	PF 8"	300	406	475	529	120	90	40	F16

■ Helical Reduction Unit

Sr. No	Models	A	B	C	D	ØE	Key	Stem Cover Thread	ØF	ØG	ØH	M	J	K	ØL	Actuator Mounting Flange
1	QTB-370/1H0	106	76.6	104.5	254	20	8x7	PF 2 1/2"	70 85	102 125	125 150	170	70	50	20	F10
2	QTB-600/1H1	121	81	119.5	291	32		PF 2 1/2"	85 100	125 140	150 175	256.5	85	70	30	F14
3	QTB-980/1H1	140	98	130	301.5	32		PF 3"	100 130	140 165	175 210	275	85	70	30	F14
4	QTB-1500/1H2	160	115	147	317	32		PF 3 1/2"	130 140	165 205	210 250	308	85	70	30	F14
5	QTB-2500/1H2	167	119	167	337	32		PF 4"	200	254	300	311	85	70	30	F14
6	QTB-3500/1H3	196	138	217.5	437	38		PF 5"	230	298	350	370	106.5	90	40	F16
7	QTB-5200/1H3	240	165.5	248	467	38		PF 6"	230	298	350	390	120	90	40	F16
8	QTB-7800/1H4	253	182.5	286	514	38		PF 6"	260	356	415	453	120	90	40	F16
9	QTB-13000-1H5	279	201	321	577	50	16x10	PF 8"	300	406	475	529	120	90	40	F16

■ FIG-(3)-Standard Gear Box with Double Reduction Unit



■ Spur Reduction Unit

Sr. No	Models	A	B	C	D	ØE	Key	Stem Cover Thread	ØF	ØG	ØH	M	J	K	ØL	Actuator Mounting Flange
1	QTB-600/2S1	121	81	119.5	343	32		PF 2 1/2"	85 100	125 140	150 175	256.5	85	70	30	F14
2	QTB-980/2S1	140	98	130	353.5	32	10x8	PF 3"	100 130	140 165	175 210	275	85	70	30	F14
3	QTB-1500/2S2	160	115	147	370	32		PF 3 1/2"	130 140	165 205	210 250	308	85	70	30	F14
4	QTB-2500/2S2	167	119	167	390	32		PF 4"	200	254	300	311	85	70	30	F14
5	QTB-3500/2S3	196	138	217.5	528	38		PF 5"	230	298	350	370	106.5	90	40	F16
6	QTB-5200/2S3	240	165.5	248	558	38	12x8	PF 6"	230	298	350	390	120	90	40	F16
7	QTB-7800/2S4	253	182.5	286	583	38		PF 6"	260	356	415	453	120	90	40	F16
8	QTB-13000/2S5	279	201	321	676	50	16x10	PF 8"	300	406	475	529	120	90	40	F16

■ Helical Reduction Unit

Sr. No	Models	A	B	C	D	ØE	Key	Stem Cover Thread	ØF	ØG	ØH	M	J	K	ØL	Actuator Mounting Flange
1	QTB-600/2H1	121	81	119.5	343	32		PF 2 1/2"	85 100	125 140	150 175	256.5	85	70	30	F14
2	QTB-980/2H1	140	98	130	353.5	32	10x8	PF 3"	100 130	140 165	175 210	275	85	70	30	F14
3	QTB-1500/2H2	160	115	147	370	32		PF 3 1/2"	130 140	165 205	210 250	308	85	70	30	F14
4	QTB-2500/2H2	167	119	167	390	32		PF 4"	200	254	300	311	85	70	30	F14
5	QTB-3500/2H3	196	138	217.5	528	38		PF 5"	230	298	350	370	106.5	90	40	F16
6	QTB-5200/2H3	240	165.5	248	558	38	12x8	PF 6"	230	298	350	390	120	90	40	F16
7	QTB-7800/2H4	253	182.5	286	583	38		PF 6"	260	356	415	453	120	90	40	F16
8	QTB-13000/2H5	279	201	321	676	50	16x10	PF 8"	300	406	475	529	120	90	40	F16

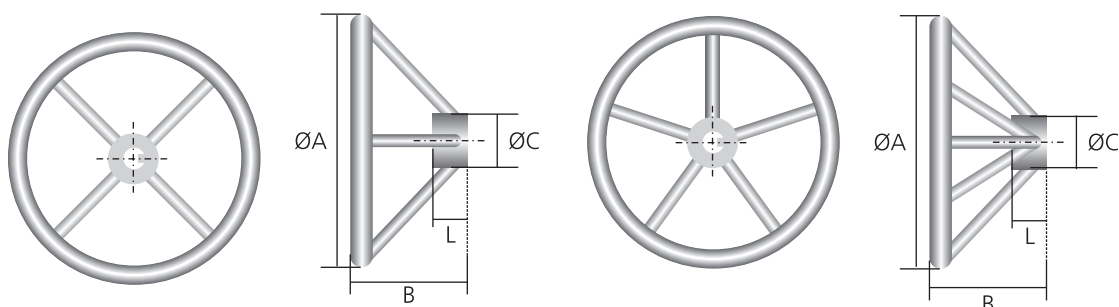
Multi-Turn Bevel Gear Box Selection Data

Sr. No	Models	Max. Output Torque		Mechanical Advantage +/-10%	Gear Ratio	Max. Thrust (Standard)		Max. Thrust (Medium)		Max.Stem Acceptance		ISO 5211 Valve Mounting	Recommended Handwheel Dia (mm)
		Nm	Ft. lbf			KN	lbf	KN	lbf	KEY	THREAD		
1	QTB-220	220	162	2.4	2.5:1	75.5	16970	166	37318	20 (8X7)	26	F10	250
2	QTB-370	370	273	2.9	3:01	113	25400	300	67443	32 (10X8)	40	F10, F12, F14	300
	QTB-370/1S0	370	273	6	6.75:1	113	25400	300	67443	32 (10X8)	40	F10, F12, F14	300
3	QTB-600	600	442	3.1	3.25:1	127	28550	335	75311	38 (12x8)	46	F12, F14, F16	400
	QTB-600/1S1	600	442	6	6.5:1	127	28550	335	75311	38 (12x8)	46	F12, F14, F16	400
	QTB-600/2S1	600	442	11.7	13:1	127	28550	335	75311	38 (12x8)	46	F12, F14, F16	400
	QTB-600/1H1	600	442	6	6.5:1	127	28550	335	75311	38 (12x8)	46	F12, F14, F16	400
	QTB-600/2H1	600	442	11.7	13:1	127	28550	335	75311	38 (12x8)	46	F12, F14, F16	400
4	QTB-980	980	723	3.3	3.5:1	141	31700	365	82055	45 (14x9)	55	F14, F16	500
	QTB-980/1S1	980	723	6.3	7:1	141	31700	365	82055	45 (14x9)	55	F14, F16	500
	QTB-980/2S1	980	723	12.6	14:1	141	31700	365	82055	45 (14x9)	55	F14, F16	500
	QTB-980/1H1	980	723	6.3	7:1	141	31700	365	82055	45 (14x9)	55	F14, F16	500
	QTB-980/2H1	980	723	12.6	14:1	141	31700	365	82055	45 (14x9)	55	F14, F16	500
5	QTB-1500	1500	1106	3.8	4:1	190	42710	450	101164	52 (16x10)	62	F16, (F20), F25	630
	QTB-1500/1S2	1500	1106	9.3	10.13:1	190	42710	450	101164	52 (16x10)	62	F16, (F20), F25	630
	QTB-1500/2S2	1500	1106	23.7	25.67:1	190	42710	450	101164	52 (16x10)	62	F16, (F20), F25	630
	QTB-1500/1H2	1500	1106	9.3	10.13:1	190	42710	450	101164	52 (16x10)	62	F16, (F20), F25	630
	QTB-1500/2H2	1500	1106	23.7	25.67:1	190	42710	450	101164	52 (16x10)	62	F16, (F20), F25	630
6	QTB-2500	2500	1844	4.8	5:1	288	64750	680	152870	60 (18x11)	72	F25, F30	710
	QTB-2500/1S2	2500	1844	11.7	12.66:1	288	64750	680	152870	60 (18x11)	72	F25, F30	710
	QTB-2500/2S2	2500	1844	29.6	32.1:1	288	64750	680	152870	60 (18x11)	72	F25, F30	710
	QTB-2500/1H2	2500	1844	11.7	12.66:1	288	64750	680	152870	60 (18x11)	72	F25, F30	710
	QTB-2500/2H2	2500	1844	29.6	32.1:1	288	64750	680	152870	60 (18x11)	72	F25, F30	710
7	QTB-3500	3500	2581	5.2	5.5:1	350	78680	735	165235	72 (20x12)	85	F25, F30	800
	QTB-3500/1S3	3500	2581	15.2	16.5:1	350	78680	735	165235	72 (20x12)	85	F25, F30	800
	QTB-3500/2S3	3500	2581	45.6	49.5:1	350	78680	735	165235	72 (20x12)	85	F25, F30	800
	QTB-3500/1H3	3500	2581	15.2	16.5:1	350	78680	735	165235	72 (20x12)	85	F25, F30	800
	QTB-3500/2H3	3500	2581	45.6	49.5:1	350	78680	735	165235	72 (20x12)	85	F25, F30	800
8	QTB-5200	5200	3835	5.7	6:1	400	89920	1000	224809	84 (22x14)	98	F30, F35	900
	QTB-5200/1S3	5200	3835	16.6	18:1	400	89920	1000	224809	84 (22x14)	98	F30, F35	900
	QTB-5200/2S3	5200	3835	49.8	54:1	400	89920	1000	224809	84 (22x14)	98	F30, F35	900
	QTB-5200/1H3	5200	3835	16.6	18:1	400	89920	1000	224809	84 (22x14)	98	F30, F35	900
	QTB-5200/2H3	5200	3835	49.8	54:1	400	89920	1000	224809	84 (22x14)	98	F30, F35	900
9	QTB-7800	7800	5753	6.2	6.5:1	510	114650	1340	301244	100 (28x16)	116	F35	900
	QTB-7800/1S4	7800	5753	21.1	22.94:1	510	114650	1340	301244	100 (28x16)	116	F35	900
	QTB-7800/2S4	7800	5753	74	80.31:1	510	114650	1340	301244	100 (28x16)	116	F35	900
	QTB-7800/1H4	7800	5753	21.1	22.94:1	510	114650	1340	301244	100 (28x16)	116	F35	900
	QTB-7800/2H4	7800	5753	74	80.31:1	510	114650	1340	301244	100 (28x16)	116	F35	900
10	QTB-13000	13000	9588	6.7	7:1	2310	519300	3100	696908	115 (32x18)	130	F40	1000
	QTB-13000/1S5	13000	9588	25.8	28:1	2310	519300	3100	696908	115 (32x18)	130	F40	1000
	QTB-13000/2S5	13000	9588	103.2	112:1	2310	519300	3100	696908	115 (32x18)	130	F40	1000
	QTB-13000/1H5	13000	9588	25.8	28:1	2310	519300	3100	696908	115 (32x18)	130	F40	1000
	QTB-13000/2H5	13000	9588	103.2	112:1	2310	519300	3100	696908	115 (32x18)	130	F40	1000

Material Specification

Sr. No	Job Description	Material Description	US Standard	IS Standard
1	Housing	Ductile Iron	A536-65-45-12	SG-450 / 10
2	Base	Ductile Iron	A536-65-45-12	SG-450 / 10
3	Bevel Gear	Carbon Steel	A576-1045 / A322 4140	AISI-1040 / AISI-4140
4	End Cap	Ductile Iron	A536-65-45-12	SG-450 / 10
5	Bevel Pinion	Carbon Steel	A576-1045 / A322 4140	AISI-1040 / AISI-4140
6	Drive Sleeve	Ductile Iron	A536-65-45-12	SG-450 / 10
7	Thrust Ball Bearing	Standard	--	--
8	O-Ring	Nitrile	--	--
9	Stem Nut	Bronze	B148 C95800	AB-2
10	Wrench Bolt	Alloy Steel	--	10.9 Gr.
11	Gasket	Cellulose Fiber 60 Gr.	--	--
12	Ball Bearing	Standard	--	--
13	Dual Lock Nuts	Carbon Steel	A576-1045	AISI-1040
14	Eye Bolt	Carbon Steel	A307 GR.B	--
15	Handwheel	Mild Steel Pipe	--	--
16	Reduction Housing	Ductile Iron	A536-65-45-12	SG-450 / 10
17	Reduction Cover	Ductile Iron	A536-65-45-12	SG-450 / 10
18	Gear	EN19	A322 4140	AISI-4140
19	Pinion	EN19	A322 4140	AISI-4140
20	Idler Gear	EN19	A322 4140	AISI-4140

Handwheel with Keyway



* Optional: Hand wheel with Dowel Hole

Sr. No	Handwheel Dia-A	Height-B	Bush OD	Dia-C	Bush Length-L	No.Of Spokes
1	250	90	70		35	4
2	300	90	70		35	5
3	400	90	70		35	5
4	500	90	70		35	5
5	630	90	70		35	5
6	710	90	70		35	5
7	800	130	80		45	6
8	900	130	80		45	6
9	900	130	80		45	6
10	1000	130	90		60	6

Ordering Information

QTE		Model			MOC-STD				VM & AM		BORE		DECIMAL		
1	2	3	4		5	6	7	8	9	10	11	12	13		
B	0	3	-		G	O	A	-	G	X	A	L	0		
Digit - 1,2,3			Digit -4	Digit - 5,6,7				Digit -8	Digit - 9,10			Digit - 11,12		Digit - 13	
Code	Model			Code	MOC - Housing, Cover Worm Wheel Worm Input Shaft			Code	Code	Mounting		Code	Bore	Code	Bore Digit
B01	QTB-220			GOA	ASTM A536 (65-45-12) AISI 1040 or EN8 AB2 AISI 1040 or EN8			-	EX	Valve Mounting-F10, Actuator Mounting-NA		***	Note 3	0	0
B02	QTB-370			HOB	ASTM A536 (65-45-12) AISI 4140 or EN19 AB2 AISI 4140 or EN19				FX	Valve Mounting-F12, Actuator Mounting-NA				1	.1
B03	QTB-600			GOF	ASTM A536 (65-45-12) AISI 1040 or EN8 AB2 SS304				GX	Valve Mounting-F14, Actuator Mounting-NA				2	.2
B04	QTB-980			HOM	ASTM A536 (65-45-12) AISI 4140 or EN19 AB2 SS316				HX	Valve Mounting-F16, Actuator Mounting-NA				3	.3
B05	QTB-1500			GOM	ASTM A536 (65-45-12) AISI 1040 or EN8 AB2 SS316				JX	Valve Mounting-F25, Actuator Mounting-NA				4	.4
B06	QTB-2500			GOH	ASTM A536 (65-45-12) AISI 1040 or EN8 AB2 SS431				KX	Valve Mounting-F30, Actuator Mounting-NA				5	.5
B07	QTB-3500			NOA	ASTM A536 (65-45-12) AISI 1040 or EN8 AB2 AISI 1040 or EN8				LX	Valve Mounting-F35, Actuator Mounting-NA				6	.6
B08	QTB-5200			KOB	ASTM A536 (65-45-12) AISI 4140 or EN19 AB2 AISI 4140 or EN19				MX	Valve Mounting-F40, Actuator Mounting-NA				7	.7
B09	QTB-7800			***	Note 1				EE	Valve Mounting-F10, Actuator Mounting-F10				8	.8
B10	QTB-13000								FG	Valve Mounting-F12, Actuator Mounting-F14				9	.9
B23	QTB-600-1S1								GE	Valve Mounting-F14, Actuator Mounting-F10					
B24	QTB-980-1S1								GG	Valve Mounting-F14, Actuator Mounting-F14					
B25	QTB-1500-1S2								HG	Valve Mounting-F16, Actuator Mounting-F14					
B26	QTB-2500-1S2								JG	Valve Mounting-F25, Actuator Mounting-F14					
B27	QTB-3500-1S3								KH	Valve Mounting-F30, Actuator Mounting-F16					
B28	QTB-5200-1S3								LH	Valve Mounting-F35, Actuator Mounting-F16					
B29	QTB-7800-1S4								MH	Valve Mounting-F40, Actuator Mounting-F16					
B30	QTB-13000-1S5								**	Note 2					
B43	QTB-600-2S1														
B44	QTB-980-2S1														
B45	QTB-1500-2S2														
B46	QTB-2500-2S2														
B47	QTB-3500-2S3														
B48	QTB-5200-2S3														
B49	QTB-7800-2S4														
B50	QTB-13000-2S5														
B63	QTB-600-1H1														
B64	QTB-980-1H1														
B65	QTB-1500-1H2														
B66	QTB-2500-1H2														
B67	QTB-3500-1H3														
B68	QTB-5200-1H3														
B69	QTB-7800-1H4														
B70	QTB-13000-1H5														
B83	QTB-600-2H1														
B84	QTB-980-2H1														
B85	QTB-1500-2H2														
B86	QTB-2500-2H2														
B87	QTB-3500-2H3														
B88	QTB-5200-2H3														
B89	QTB-7800-2H4														
B90	QTB-13000-2H5														
T01	QTB-220M														
T02	QTB-370M														
T03	QTB-600M														
T04	QTB-980M														
T05	QTB-1500M														
T06	QTB-2500M														
T07	QTB-3500M														
T08	QTB-5200M														
T09	QTB-7800M														
T10	QTB-13000M														
T23	QTB-600M-1S1														
T24	QTB-980M-1S1														
T25	QTB-1500M-1S2														
T26	QTB-2500M-1S2														
T27	QTB-3500M-1S3														
T28	QTB-5200M-1S3														
T29	QTB-7800M-1S4														
T30	QTB-13000M-1S5														
T43	QTB-600M-2S1														
T44	QTB-980M-2S1														
T45	QTB-1500M-2S2														
T46	QTB-2500M-2S2														
T47	QTB-3500M-2S3														
T48	QTB-5200M-2S3														
T49	QTB-7800M-2S4														
T50	QTB-13000M-2S5														
T63	QTB-600M-1H1														
T64	QTB-980M-1H1														
T65	QTB-1500M-1H2														
T66	QTB-2500M-1H2														
T67	QTB-3500M-1H3														
T68	QTB-5200M-1H3														
T69	QTB-7800M-1H4														
T70	QTB-13000M-1H5														
T83	QTB-600M-2H1														
T84	QTB-980M-2H1														
T85	QTB-1500M-2H2														
T86	QTB-2500M-2H2														
T87	QTB-3500M-2H3														
T88	QTB-5200M-2H3														
T89	QTB-7800M-2H4														
T90	QTB-13000M-2H5														

Note 1	For more MOC combinations, please contact Q-Tork
Note 2	For multiple Gearbox mounting PCD, please contact Q-Tork
Note 3	Detailed coding is mentioned on Page 62 of this Catalogue
Note 4	For additional type and size, Please contact Q-Tork
Note 5	For additional accessories, please specify
Note 6	For more specials, please specify

Note 1 For more MOC combinations, please contact Q-Tork

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Note 5 For additional accessories, please specify

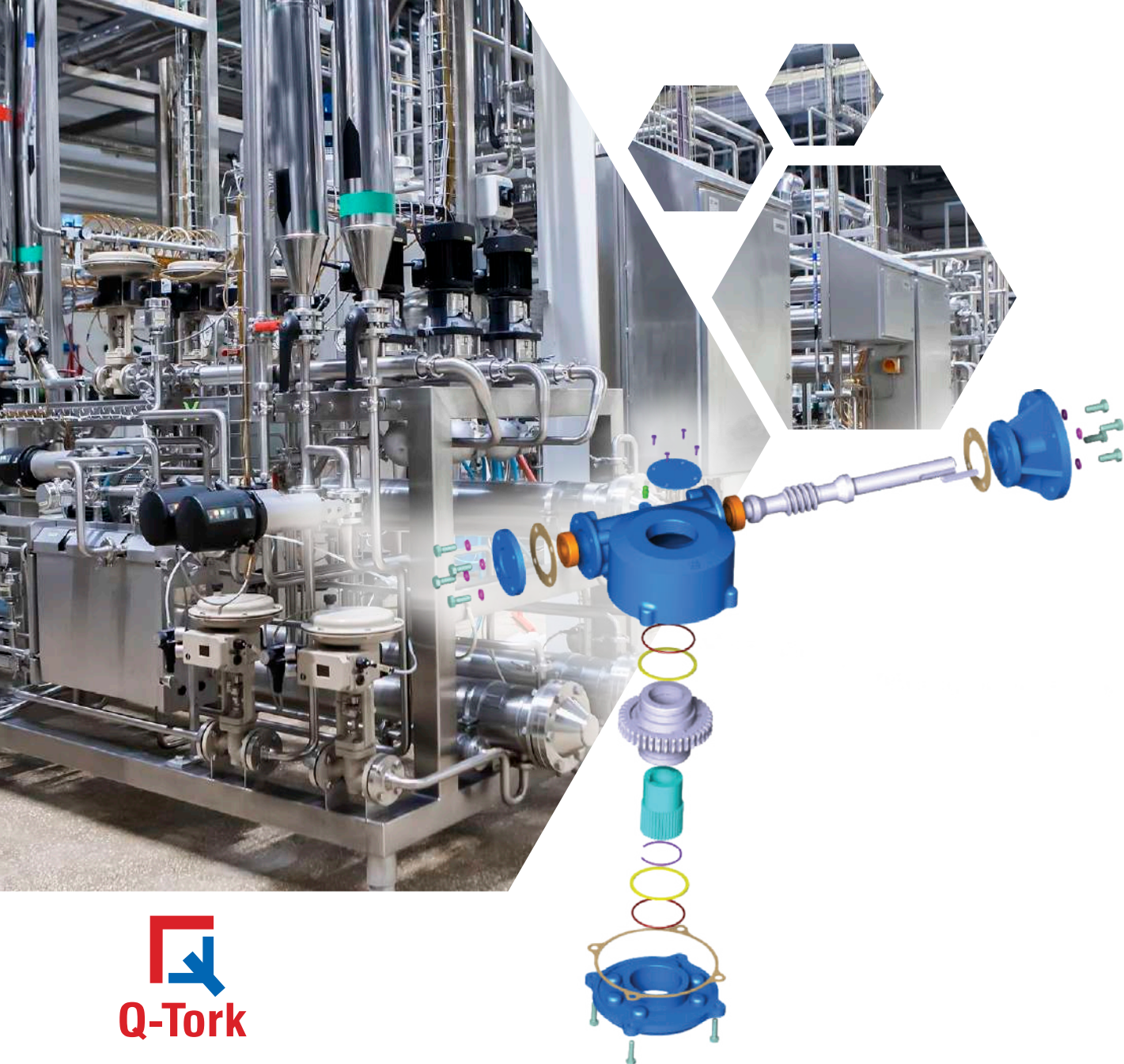
Note 6 For more specials, please specify

ORIENTATION			HAND WHEEL		ACCESSORIES		IP / LA		SPECIAL		
14	15	16	17	18	19	20	21	22	23	24	25
X	X	-	K	F	X	X	1	-	X	X	X
Digit - 14,15		Digit - 16	Digit - 17,18		Digit - 19,20		Digit - 21		Digit-22	Digit - 23,24,25	
Code	Orientation	Code	Code	Handwheel / Chain Wheel	Code	Accessories	Code	IP rating / Padlock	Code	Code	Special
XX	NA	-	KB	MS Fabricated H/W:- Dia 200mm Key Type	XX	No accessories	2	Enclosure type:- IP67 Without Pad Lock	-	AAH	Hardware SS316, Gear box RAL5015
			KE	MS Fabricated H/W:- Dia 350mm Key Type	**	Note 4	3	Enclosure type:- IP68 Without Pad Lock		AAS	Gear Box RAL 9005
			KH	MS Fabricated H/W:- Dia 500mm Key Type			4	Enclosure type:- Submersible Without Pad Lock		AAU	Hardware SS316, Gear box RAL9005
			KK	MS Fabricated H/W:- Dia 600mm Key Type			6	Enclosure type:- IP67 With Pad Lock		AAY	Gear box RAL5015
			KN	MS Fabricated H/W:- Dia 700mm Key Type			7	Enclosure type:- IP68 With Pad Lock		ABE	Hardware SS304
			XX	Without Handwheel			8	Enclosure type:- Submersible With Pad Lock		ABF	Hardware SS316
			KM	MS Fabricated H/W:- Dia 650mm Key Type						ABJ	Gear box unpainted
			XX	Without Handwheel						ACJ	Low Temperature Grease
			**	Note 4						XXX	without any specials
										***	Note 5

Build Your Code :

Build Your Code :

QTB	MODEL			MOC-STD.				VM		BORE		DECIMAL	ORIENTATION			HAND WHEEL		ACCESSORIES		IP / LA		SPECIAL			
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
			-				-								-						-				



518 Series | Multi-turn Worm Gear Operators with Spline for Electrical and Manual Operations

Q-Tork 518 series gear boxes are designed for both Electrical and Manual operation. This type of gear boxes are designed for 360 degree rotation without end stop.

Application

518 Series multi-turn worm gear boxes are designed to operate Sluice valves, Balancing valves and Dampers

Features & Benefits

- Spline bush provides accurate positioning - valve stem and worm wheel
- Manual $\rightleftharpoons$ Electrical interchangeability
- Cast iron housing and cover
- Flange is provided for mounting of electric actuator
- Temperature range -30°C to +120°C
- Light weight with rugged design
- Maintenance free
- Environmental friendly high performance grease

Optional

- Housing materials - Carbon steel,
- Stainless steel, Ductile iron
- Submersibility to IP67
- Worm wheel - Al bronze
- Stainless steel-Input shaft, handwheel and enclosed screws
- Can be made suitable for 90° and 180° operation

■ Gear Box Selection Data

Sr. No	Model	Max. Output Torque Manual (Nm)	Max. Output Torque Electrical (Nm)	Gear Ratio	MA ± 10 %	ISO 5211 Valve Mounting	ISO 5211 Actuator Mounting	Max Stem Acceptance Dia (mm)
1	QMW-0.5	500	360	21:1	6.6	F07, F10	F07, F10	28
2	QMW-01	1000	700	39:1	12.3	F10, F12	F07, F10	36
3	QMW-02	1700	1200	40:1	13.7	F12, F14	F10	48
4	QMW-04	3400	2400	40:1	13.4	F14, 16	F10	60
5	QMW-07	6700	4800	52:1	16.9	F16, F25	F10	72
6	QMW-14	13500	9600	54:1	17.1	F25, F30	F12	98
7	QMW-28	27500	19200	53:1	17	F30, F35	F14	130
8	QMW-36	36000	25600	57:1	18	F30, F40	F14	150
9	QMW-56	56000	38400	52:1	16	F40	F16	160
10	QMW-76	76000	54000	53:1	16.7	F40	F16	180
11	QMW-108	108000	75600	53:1	16.4	F48	F25	200
12	QMW-143	143000	102000	54:1	17	F48	F25	220

■ Material Specifications

Sr. No	Job Description	Material Description	Standards
1	Housing	Cast Iron-FG 220	IS 210/ASTM A48 /40B/GG-20
2	Cover	Cast Iron-FG 220	IS 210/ASTM A48 /40B/GG-20
3	Worm Wheel	SG Iron-600/3	IS 1865 / ASTM A536 / GGG-60
4	Worm Shaft	EN-19	C55 / SAE-4140 / 42CrMo4 / ASTM A-29 / A-29M
5	Spline Bush	SG Iron-600/3	IS 1865 / ASTM A536 / GGG-60
6	Bearing	Standard	—
7	O-ring	Nitrile	—
8	Handwheel	Mild Steel Pipe	—
9	Drive Shaft	EN-19	C55 / SAE-4140 / 42CrMo4 / ASTM A-29 / A-29M

Ordering Information

QMW	Model			MOC-STD					VM & AM		BORE		DECIMAL		
1	2	3	4	5	6	7	8	9	10	11	12	13			
W	0	1	-	M	B	B	-	D	E	S	H	0			
Digit - 1,2,3			Digit - 4	Digit - 5,6,7				Digit - 8	Digit - 9,10		Digit - 11,12		Digit - 13		
Code	Model	-	Code	MOC - Housing, Cover Worm Wheel Worm Input Shaft				Code	Code	Mounting		Code	Bore	Code	Bore Digit
W01	QMW-0.5		MBB	ASTM A48 Class 35B	ASTM A536 (80-55-06)	AISI 4140 or EN19	AISI 4140 or EN19	-	DX	Valve Mounting-F07, Actuator Mounting-NA		***	Note 3	0	0
W02	QMW-01		***	Note 1					EX	Valve Mounting-F10, Actuator Mounting-NA				1	.1
W03	QMW-02								FX	Valve Mounting-F12, Actuator Mounting-NA				2	.2
W04	QMW-04								GX	Valve Mounting-F14, Actuator Mounting-NA				3	.3
W05	QMW-07								HX	Valve Mounting-F16, Actuator Mounting-NA				4	.4
W06	QMW-14								JX	Valve Mounting-F25, Actuator Mounting-NA				5	.5
W07	QMW-28								KX	Valve Mounting-F30, Actuator Mounting-NA				6	.6
W08	QMW-36								LX	Valve Mounting-F35, Actuator Mounting-NA				7	.7
W09	QMW-56								MX	Valve Mounting-F40, Actuator Mounting-NA				8	.8
W10	QMW-76								NX	Valve Mounting-F48, Actuator Mounting-NA				9	.9
W11	QMW-108								DD	Valve Mounting-F07, Actuator Mounting-F07					
W12	QMW-143								DE	Valve Mounting-F07, Actuator Mounting-F10					
									ED	Valve Mounting-F10, Actuator Mounting-F07					
									EE	Valve Mounting-F10, Actuator Mounting-F10					
									FD	Valve Mounting-F12, Actuator Mounting-F07					
									FE	Valve Mounting-F12, Actuator Mounting-F10					
									GE	Valve Mounting-F14, Actuator Mounting-F10					
									HE	Valve Mounting-F16, Actuator Mounting-F10					
									JE	Valve Mounting-F25, Actuator Mounting-F10					
									KG	Valve Mounting-F30, Actuator Mounting-F14					
									LG	Valve Mounting-F35, Actuator Mounting-F14					
									MG	Valve Mounting-F40, Actuator Mounting-F14					
									MH	Valve Mounting-F40, Actuator Mounting-F16					
									NJ	Valve Mounting-F48, Actuator Mounting-F25					
								**	Note 2						

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Note 2	For multiple Gearbox mounting PCD, please contact Q-Tork
Note 3	Detailed coding is mentioned on Page 62 of this Catalogue
Note 4	For additional accessories, please specify
Note 5	For more specials, please specify

ORIENTATION			HAND WHEEL			ACCESSORIES		IP / LA				SPECIAL		
14	15	16	17		18	19	20	21			22	23	24	25
X	X	-	X		X	X	X	1			-	X	X	X
Digit - 14,15		Digit - 16	Digit - 17,18			Digit - 19,20		Digit - 21			Digit-22	Digit - 23,24,25		
Code	Orientation	Code	Code	Handwheel / Chain Wheel		Code	Accessories	Code	IP rating / Padlock			Code	Code	Special
XX	NA	-	KB	MS Fabricated H/W/- Dia 200mm Key Type		XX	No accessories	2	Enclosure type:- IP67 Without Pad Lock			-	AAH	Hardware SS316, Gear box RAL5015
			KE	MS Fabricated H/W/- Dia 350mm Key Type		**	Note 4	3	Enclosure type:- IP68 Without Pad Lock				AAS	Gear Box RAL 9005
			KH	MS Fabricated H/W/- Dia 500mm Key Type				4	Enclosure type:- Submersible Without Pad Lock				AAU	Hardware SS316, Gear box RAL9005
			KK	MS Fabricated H/W/- Dia 600mm Key Type				6	Enclosure type:- IP67 With Pad Lock				AAY	Gear box RAL5015
			KN	MS Fabricated H/W/- Dia 700mm Key Type				7	Enclosure type:- IP68 With Pad Lock				ABE	Hardware SS304
			XX	Without Handwheel				8	Enclosure type:- Submersible With Pad Lock				ABF	Hardware SS316
								7	Enclosure type:- IP68 With Pad Lock				ABJ	Gear box unpainted
								8	Enclosure type:- Submersible With Pad Lock				ACJ	Low Temperature Grease
												XXX	without any specials	
												***	Note 5	

Build Your Code :

Blank Year Code :																										
QMW	MODEL			MOC-STD.W				VM			BORE		DECIMAL	ORIENTATION			HAND WHEEL		ACCESSORIES			IP / LA		SPECIAL		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
			-				-								-						-					

Bore Information

* Use this page in conjunction with page numbers 16, 24, 30, 40, 48, 56, 60

Digit - 11,12	
Code	Bore
AA	10
AB	11
AC	12
AD	13
AE	14
AF	15
AG	16
AH	17
AJ	18
AK	19
AL	20
AM	21
AN	22
AP	23
AQ	24
AR	25
AS	26
AT	27
AU	28
AV	29
AW	30
AY	31
AZ	32
BA	33
BB	34
BC	35
BD	36
BE	37
BF	38
BG	39
BH	40
BJ	41
BK	42
BL	43
BM	44
BN	45
BP	46
BQ	47
BR	48
BS	49
BT	50
BU	51
BV	52
BW	53
BY	54
BZ	55
CA	56
CB	57
CC	58
CD	59
CE	60
CF	61
CG	62
CH	63
CJ	64
CK	65
CL	66
CM	67
CN	68
CP	69
CQ	70
CR	71
CS	72
CT	73
CU	74
CV	75
CW	76
CY	77
CZ	78
EA	79
EB	80
EC	81
ED	82
EE	83
EF	84
EG	85
EH	86
EJ	87
EK	88
EL	89
EM	90

Digit - 11,12	
Code	Bore
EN	91
EP	92
EQ	93
ER	94
ES	95
ET	96
EU	97
EV	98
EW	99
EY	100
EZ	101
FA	102
FB	103
FC	104
FD	105
FE	106
FF	107
FG	108
FH	109
FJ	110
FK	111
FL	112
FM	113
FN	114
FP	115
FQ	116
FR	117
FS	118
FT	119
FU	120
FV	121
FW	122
FY	123
FZ	124
GA	125
GB	126
GC	127
GD	128
GE	129
GF	130
GG	131
GH	132
GJ	133
GK	134
GL	135
GM	136
GN	137
GP	138
GQ	139
GR	140
GS	141
GT	142
GU	143
GV	144
GW	145
GY	146
GZ	147
HA	148
HB	149
HC	150
HD	151
HE	152
HF	153
HG	154
HH	155
HJ	156
HK	157
HL	158
HM	159
HN	160
HP	161
HQ	162
HR	163
HS	164
HT	165
HU	166
HV	167
HW	168

Digit - 11,12	
Code	Bore
HY	169
HZ	170
JA	171
JB	172
JC	173
JD	174
JE	175
JF	176
JG	177
JH	178
JJ	179
JK	180
JL	181
JM	182
JN	183
JP	184
JQ	185
JR	186
JS	187
JT	188
JU	189
JV	190
JW	191
JY	192
JZ	193
KA	194
KB	195
KC	196
KD	197
KE	198
KF	199
KG	200
KH	201
KJ	202
KK	203
KL	204
KM	205
KN	206
KP	207
KQ	208
KR	209
KS	210
KT	211
KU	212
KV	213
KW	214
KY	215
KZ	216
LA	217
LB	218
LC	219
LD	220
LE	221
LF	222
LG	223
LH	224
LJ	225
LK	226
LL	227
LM	228
LN	229
LP	230
LQ	231
LR	232
LS	233
LT	234
LU	235
LV	236
LW	237
LY	238
LZ	239
MA	240
MB	241
MC	242

Digit - 11,12	
Code	Bore
MD	243
ME	244
MF	245
MG	246
MH	247
MJ	248
MK	249
ML	250
MM	251
MN	252
MP	253
MQ	254
MR	255
MS	256
MT	257
MU	258
MV	259
MW	260
MY	261
MZ	262
NA	263
NB	264
NC	265
ND	266
NE	267
NF	268
NG	269
NH	270
NJ	271
NK	272
NL	273
NM	274
NN	275
NP	276
NQ	277
NR	278
NS	279
NT	280
NU	281
NV	282
NW	283
NY	284
NZ	285
PA	286
PB	287
PC	288
PD	289
PE	290
PF	291
PG	292
PH	293
PJ	294
PK	295
PL	296
PM	297
PN	298
PP	299
PQ	300
PR	301
PS	302
PT	303
PU	304
PV	305
PW	306
PY	307
PZ	308

Digit - 11,12	
Code	Double D
DA	16x11
DB	19x13
DC	22x16
DD	25x19
DE	30x22
DF	12x9
DG	14x11
DH	18x14
DJ	22x17
DK	28x22
DL	36x27
DM	48x36
DN	60x46
DP	72x55
DQ	98x75
DR	14x10
DS	18x13
DT	28x20
DU	25x20

Digit - 11,12	
Code	Square
SA	11x11
SB	14x14
SC	17x17
SD	19x19
SE	22x22
SF	24x24
SG	25x25
SH	27x27
SJ	36x36
SK	46x46
SL	55x55
SM	10x10
SN	12x12
SP	13x13
SQ	15x15
SR	16x16
SS	18x18
ST	20x20
SU	21x21
SV	23x23
SW	26x26
SY	28x28
SZ	29x29
S1	30x30
S2	31x31
S3	32x32
S4	33x33
S5	34x34
S6	35x35
S7	37x37
S8	38x38
S9	39x39
TA	9x9
TB	40x40
TC	19x14
XX	Pilot Bore
***	Note 3

* Dimensions for Keyways as per standard IS2048:1983

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- Quarter-turn Worm Gear Operators for Electrical Actuation • Multi-turn Bevel Gear Operators
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